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ACADEMIC INTERNSHIP REPORT

**THEME: CONCEPTION AND REALISATION OF A DIGITAL
HEALTH SUPPORT AND GENOTYPE COMPATIBILITY
PLATFORM FOR SICKLE CELL PREVENTION IN
CAMEROON**

Internship carried out from the 1st of July to the 30th of September 2025 at the
company **REALIZE** as part of my pursuit to obtain a
Higher Technician Diploma (HTD) in Computer Science

Option: Software Engineering

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CONCEPTION AND REALISATION OF A DIGITAL HEALTH
SUPPORT AND GENOTYPE COMPATIBILITY PLATFORM FOR
SICKLE CELL PREVENTION IN CAMEROON



DEDICATION

THIS BOOK IS DEDICATED TO THE METOMO FAMILY

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GLOSSARY

- **2TUP:** Two-Track Unified Process
- **AICS:** African Institute of Computer Sciences
- **AI:** Artificial Intelligence
- **API:** Application Programming Interface
- **IAI :** Institut Africain de L'Informatique
- **ICT:** Information and Communication Technologies
- **MERISE :** Méthode d'Étude et de Réalisation Informatique pour les Systèmes d'Entreprise
- **MOMO:** Mobile Money
- **MTN:** Mobile Telecommunications Network
- **OM :** Orange Money
- **TIC :** Technologies de l'Information et de la Communication
- **UI:** User Interface
- **UML:** Unified Modeling Language

ABSTRACT

The role of Information and Communication Technologies (ICT) in fostering development remains a topic of debate. In today's information-driven world, ICTs play a decisive role in Africa's growth and are increasingly recognized as essential for progress. For this reason, most companies are embracing this trend to increase their business activity and ensure the sustainability of their various services. Our internship, which aimed to contribute to this major ICT-driven development, took place at REALIZE, a Cameroon-based non-governmental tech start-up. This provided us with the opportunity to work on several tasks, including the " **CONCEPTION AND REALISATION OF A DIGITAL HEALTH SUPPORT AND GENOTYPE COMPATIBILITY PLATFORM FOR SICKLE CELL PREVENTION IN CAMEROON.**" This application provides individuals, couples, and families with reliable digital tools to better understand sickle cell disease, check genotype compatibility before marriage, and access health support resources using only a smartphone or any internet-connected device. With the knowledge gained at AICS Cameroon, we successfully developed the application. Using UML diagrams, we were able to model the system and better understand its structure. This report outlines the stages involved in designing, developing, and implementing the application.

Keywords:

- Genotype
- Technologies
- Information Technology
- Realize
- Sickle Cell

RESUME

Le rôle des Technologies de l'Information et de la Communication (TIC) dans la promotion du développement demeure un sujet de débat. Dans le monde actuel, axé sur l'information, les TIC jouent un rôle décisif dans la croissance de l'Afrique et sont de plus en plus reconnues comme essentielles au progrès. Pour cette raison, la plupart des entreprises adoptent cette tendance afin d'accroître leurs activités et d'assurer la pérennité de leurs différents services. Notre stage, qui visait à contribuer à ce développement majeur porté par les TIC, s'est déroulé au sein de **REALIZE**, une start-up technologique camerounaise à but non lucratif. Cela nous a donné l'opportunité de travailler sur plusieurs tâches, notamment le projet intitulé « **CONCEPTION ET RÉALISATION D'UNE PLATEFORME NUMÉRIQUE DE SOUTIEN SANITAIRE ET DE COMPATIBILITÉ DE GÉNOTYPE POUR LA PRÉVENTION DE LA DRÉPANOCYTOSE AU CAMEROUN** ». Cette application met à la disposition des individus, des couples et des familles des outils numériques fiables pour mieux comprendre la drépanocytose, vérifier la compatibilité des génotypes avant le mariage et accéder à des ressources de soutien sanitaire à l'aide d'un simple smartphone ou de tout autre appareil connecté à Internet. Grâce aux connaissances acquises à l'AICS Cameroun, nous avons pu développer avec succès l'application. À l'aide des diagrammes UML, nous avons pu modéliser le système et mieux comprendre sa structure. Ce rapport présente les étapes impliquées dans la conception, le développement et la mise en œuvre de l'application.

Mots-clés :

- Drépanocytose
- Génotype
- Realize
- Stage
- Technologies de l'Information



GENERAL INTRODUCTION

Despite various challenges, the field of computer science has continuously evolved and expanded its impact. Today, individuals worldwide face numerous difficulties that require effective technological solutions tailored to their needs. Organizations increasingly depend on professionals with strong training in Information Technology and Communication to address these demands. In response to these challenges, the African Institute of Computer Sciences (AICS), a leading institution in the field, has established a three-month internship program for Level 2 students. This program is designed to bridge theoretical knowledge and practical skills by engaging students in real-world projects, enhancing their competencies while supporting organizations in overcoming obstacles and fostering growth. The selected organization for this internship was REALIZE, a software development company committed to sustainable development. The project focused on the "Conception and realization of a digital health support and genotype compatibility platform for sickle cell prevention in Cameroon," which was structured into eight main phases to ensure comprehensive development and deployment.

To ease understanding, we divided this document into eight (8) parts. These are:

1. Insertion Phase: Here we shall talk about our host company, their services, mission, as well as our first two weeks at our host company.
2. Existing System: Here, we shall present the already present system in place, that is the one used for consultation and follow-up purposes.
3. Specification Book: In this book, we specify the needs of the user, taking into consideration the time and cost of the project.
4. Analysis Document: Here, we shall present the analysis method chosen, together with the presentation of all the diagrams used for the analysis of this project.
5. Conception phase: This presents the generic and detailed conception of the project to bring out real-world constituents.
6. The Realization phase: This phase will permit us to visualize the implementation process of the solution.
7. Test of functionalities: In this phase, we shall present to you the different functionalities or modules of our application and how they work.
8. The User Guide: This elaborates on all conditions necessary to use the application and how to use it.



PART ONE: INSERTION PHASE



Preamble

The inserted document forms part of my internship report, which details the structure and objectives of the organization where I completed my internship. It presents an overview of my internship environment and delineates the different stages I underwent to successfully assimilate into the company

Content:

INTRODUCTION

- I. WELCOMING AND INTEGRATION
- II. GENERAL PRESENTATION OF REALIZE
- III. HARDWARE AND SOFTWARE RESOURCES
- IV. BRIEF PRESENTATION OF THE PROJECT

CONCLUSION



CONCEPTION AND REALISATION OF A DIGITAL HEALTH SUPPORT AND GENOTYPE COMPATIBILITY PLATFORM FOR SICKLE CELL PREVENTION IN CAMEROON



INTRODUCTION

Professional integration is a process that enables individuals or groups to enter the job market under conditions that enhance their chances of securing employment. During our academic internship, we had the opportunity to be intern at REALIZE for three months, with the integration phase spanning two weeks. This section of the internship report outlines the structure and operations of the organization, the experience of the student during the reception period, and the research topic pursued during this time.

I. WELCOMING AND INTEGRATION

The welcoming and integration event took place on the 1st of July at 8:15 am. Upon arrival, we were greeted by Mr. TSEMBOM PERCY, the senior developer, who delivered an inspiring welcoming speech. Having interned at REALIZE last year, this moment felt familiar, though it was refreshing to reconnect with the team. We proceeded with a presentation and a meeting focused on discussing the challenges we faced during the coding test, which was a prerequisite for our internship at the company. The meeting also included an overview of the enterprise's structure, its internal policies, and the specific workstations assigned to each intern. Given my previous experience, I was able to quickly reorient myself within the company's environment. Mr. Tanue concluded by encouraging us to strive for excellence and offered his support for any issues we might encounter.

After settling into our designated workstations, we were promptly assigned our first task for the day. During the first week, we conducted a verification of concepts using fundamental HTML and CSS to ensure everyone was on an even playing field before progressing. We then engaged in discussions about project ideas, where we were encouraged to propose our own. My previous experience made it easier to contribute and refine these ideas. Subsequently, we were tasked with researching these ideas, as the enterprise valued our initiative and did not impose any themes on us.

In the second week, we focused on JavaScript concept verification exercises to assess our understanding and reasoning skills. We held a detailed discussion with our professional supervisor about the specification book and its various components, where he outlined what was expected of us in each segment. We also continued developing our project ideas into a concrete plan, benefiting from my previous familiarity with the company's expectations and processes.

II. GENERAL PRESENTATION OF REALIZE

i. Background

Realize is a Cameroon based non-governmental tech start-up institute founded by NDELOKAKEH Daniel in 2022 which proposes IT solutions and empowers the use of new technologies in Cameroon

ii. Missions, Visions and Activities of Realize

1. Missions:

The mission of Realize mainly relates to the empowerment and the perpetuation of Technology use. These missions include:

- Design and realize mobile-oriented software for companies and individuals;
- Assist newborn companies for quick growth using computer sciences;
- Provide training and certifications to improve qualified human resources in many fields of study;

Take an active part in the sustainable development of the world through innovative solutions and virtual reality.

2. Vision:

At Realize, we believe that the true potential of every organization and individual can be unlocked through the harmonious fusion of analysis, development, realization, and data analytics. Our vision propels us forward, guiding us to make a lasting impact on the digital landscape and shape a future where innovation knows no bounds."

3. Activities:

The activities of Realize range from computer sciences, engineering, and training. We can outline the following:

- Conception realization, and hosting of websites;
- Software development and maintenance;
- Training in Software related fields;
- Conception and realization of multimedia;
- IT consulting and innovation
- IT support

1. Localization



Figure 1 Realise localisation

III. ORGANIZATION OF REALIZE

A. Administrative organization

Realize is administratively organized as follows:

a. The General management

This is the highest level of the company, which takes care of the following:

- Proper functioning of each department;
- Define project strategies;
- Provide leadership and guidance to the company's employees.
- Makes critical decision that could affect the company's operations or reputation
- Severe as a point of contact for key stakeholders such as investors etc.

b. Human resource department

This department is in charge of the following:

- Recruitment and Hiring
- Acts as a liaison between employees and management;
- Manages employee compensation and benefits programs;
- Develops and implements company policies and procedures, ensuring compliance with employment laws and regulations.
- Manages administrative tasks related to employee records, data management, and HR systems

c. Communication Department

This department is in charge of the following:

- Handles public relations activities, which involve managing the company's reputation and image;
- Responsible for crafting and delivering messages to external stakeholders such as customers etc.;
- It fosters effective communication within the company;
- They create and manages content across different platforms and channels;

d. Department of Financial Affairs

This department is in charge of the following:

- Responsible for developing and managing the company's financial plans and budgets;
- Prepare and presents accurate and timely financial reports to management, stakeholders, and regulatory authorities;
- Responsible for managing cost and expenses within the organization;
- Ensures compliance with tax laws and regulations. Etc.

e. Technical department

This department is in charge of the following:

- Responsible for managing the company's technological infrastructure, including networks, hardware;

- Play a role in managing and maintaining the company's data;
- Responsible for implementing and maintaining cyber security measures to protect the company's digital assets from potential threats;
- Interacts with technology vendors and manages relationships with external service providers. Etc.

f. Software Engineering Department

This department is in charge of the following:

- Primarily responsible for developing software applications and systems;
- Responsible for maintaining and supporting software application throughout their lifecycle;
- Evaluation and realization of projects etc.

B. FUNCTIONAL ORGANIZATION OF REALIZE

The functional branch of Realize is organized as follows

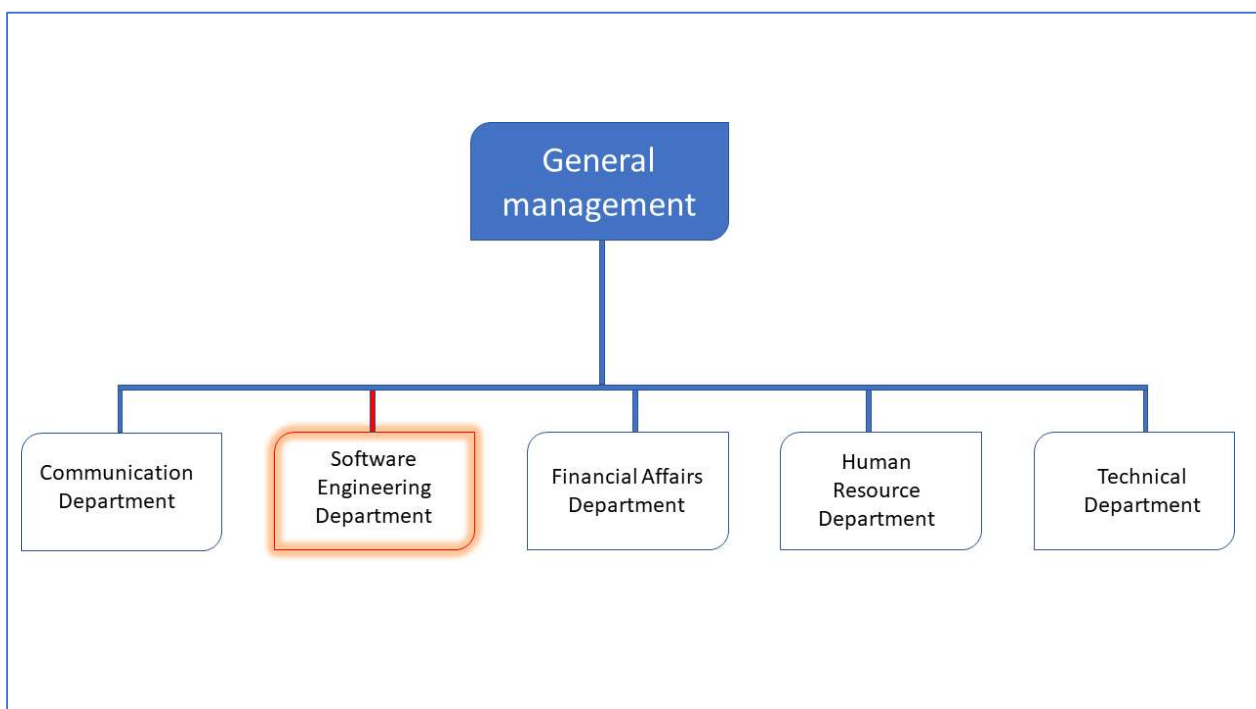


Figure 2 functional branch of realize

IV. HARDWARE AND SOFTWARE RESOURCES

A. HARDWARE RESOURCES

Table 1 hardware resource

No	DESIGNATION	QUANTITY	CHARECTERISTICS	OBSERVATION
1	MacBook Pro	1	APPLE	Good
2	Modem	1	CAMTEL	Good
3	LCD Screen	1	HP	Excellent
4	Training Equipment and office furniture	/	/	/

B. SOFTWARE RESOURCES

Table 2 software resources

DESIGNATION	SOFTWARE
Operating System	Windows 10, Mac OS, parrot Linux
Design tools	Photoshop
Integrated development environment (IDE)	Visual studio code
Text editor	Sublime text, notepad++
Database management system (DBMS)	MongoDB, PostgreSQL, MySQL
Web browser	Google chrome, Microsoft edge
Document editor	Microsoft office word
Presentation	Microsoft office PowerPoint

V. BRIEF PRESENTATION OF PROJECT

At the end of the initial two-week orientation period of my internship at Realize, we were assigned the project titled "CONCEPTION AND REALISATION OF A DIGITAL HEALTH SUPPORT AND GENOTYPE COMPATIBILITY PLATFORM FOR SICKLE CELL PREVENTION IN CAMEROON." This project involves developing a comprehensive healthcare mobile application designed to support sickle cell patients and their families. The platform empowers users with tools for health management, crisis guidance, and genotype compatibility analysis to assess the risk of having a child affected by sickle cell disease. Using modern technologies such as MongoDB, React Native, and Node.js, and applying system analysis methodologies like UML and the 2TUP process, the application aims to simplify the understanding and management of sickle cell risks. Its primary objective is to provide an accessible, user-friendly, and reliable tool for families in Cameroon, ultimately contributing to reducing the prevalence of the disease through informed decision-making and early prevention.

CONCLUSION

Our insertion phase at REALIZE lasted for two weeks. During this time, we discovered the remarkable environment we were in, which encourages hard work, perseverance, and collaboration; qualities that many students seek. We learned about the company's missions and activities, and we had the opportunity to meet and connect with the staff and other interns at REALIZE. As a culmination of this phase, we were assigned the theme “CONCEPTION AND REALISATION OF A DIGITAL HEALTH SUPPORT AND GENOTYPE COMPATIBILITY PLATFORM FOR SICKLE CELL PREVENTION IN CAMEROON

”

PART TWO: TECHNICAL PHASE

Preamble

This chapter will present and elucidate our central theme, which is the primary emphasis and objective of our project. We shall also expound on the predicament that prompted us to devise our proposed solution, and detail the characteristics and operations of the software we have planned to construct.

Content:

INTRODUCTION

- THEME PRESENTATION
- STUDY OF THE EXISTING SYSTEM
- LIMITATION OF THE EXISTING SYSTEM
- PROBLEMATIC
- PROPOSED SOLUTION

CONCLUSION

FILE ONE: EXISTING SYSTEM

Preamble

This chapter will present and elucidate our central theme, which is the primary emphasis and objective of our project. We shall also expound on the predicament that prompted us to devise our proposed solution, and detail the characteristics and operations of the software we have planned to construct.

Content:

INTRODUCTION

- I. PRESENTATION OF THE THEME
- II. STUDY OF THE EXISTING SYSTEM
- III. LIMITATIONS OF THE EXISTING SYSTEM
- IV. PROBLEMATIC
- V. PROPOSED SOLUTIONS

CONCLUSION



INTRODUCTION

In this part of our report, we will clearly explain our theme to provide a solid understanding of the subject. We will also take time to study the current system in detail and carefully identify both its strengths and weaknesses. This analysis will help us highlight what is working well and what still needs improvement. Based on the findings, we will then present the problem statement, which represents the main challenge or gap that the current system faces. By clearly stating the problem, we can demonstrate why a new approach is necessary. Ultimately, we will propose a well-considered solution that directly addresses this issue and enhances the system's overall performance.

I. PRESENTATION OF THE THEME

The theme of this project is the **CONCEPTION AND REALISATION OF A DIGITAL HEALTH SUPPORT AND GENOTYPE COMPATIBILITY PLATFORM FOR SICKLE CELL PREVENTION IN CAMEROON**, developed during an internship at Realize. This mobile application aims to empower families affected by sickle cell disease by providing an accessible platform that supports health management and offers genotype compatibility analysis to assess the risk of having a child with sickle cell disease. The app simplifies risk assessment by enabling parents to input genotype information and receive clear, accurate predictions regarding their child's sickle cell risk. It caters to diverse user needs by offering both detailed genetic risk evaluations and straightforward educational guidance to support informed family planning in Cameroon.

Healthcare providers can integrate their services within the platform, offering genotype testing and genetic counseling tailored to local contexts. This integration allows providers to expand their reach while enabling families to select the appropriate level of healthcare support. To ensure reliability and trust, the app includes a verification process for healthcare providers, guaranteeing that only accredited and trustworthy professionals are available to users. The system design follows rigorous methodologies including UML and the 2TUP process, ensuring a secure, user-friendly, and robust platform that contributes to reducing the prevalence of sickle cell disease in Cameroon.

II. STUDY OF THE EXISTING SYSTEM

In the current system, the assessment of sickle cell risk in Cameroon is often informal and fragmented. Many families rely on limited information from community sources or personal networks to understand their sickle cell status, as these channels appear more accessible. However, this informal approach lacks the accuracy and reliability provided by professional healthcare services, leading to insufficient awareness, delayed genotype testing, and increased anxiety about the risk of having a child affected by sickle cell disease.

For those seeking professional care, options remain scarce. Only a few healthcare facilities in Cameroon offer accessible genotype testing or genetic counseling, and these services rarely have a strong online presence to provide easily accessible sickle cell risk information. Such resources mainly serve urban or specialized populations, leaving many families without

convenient access to reliable healthcare support. The absence of a centralized digital platform means that parents must manually contact clinics or hospitals for testing, counseling, or risk evaluation, which is time-consuming and lacks standardization.

Additionally, the current system suffers from a lack of transparency. Without integrated educational materials or verified healthcare provider information, families struggle to evaluate the quality and trustworthiness of available services. The limited online visibility of sickle cell testing centers worsens this issue, forcing parents to depend on word-of-mouth or outdated data. This fragmented and inaccessible landscape underscores the urgent need for a unified, user-friendly, and trustworthy platform. The mobile application developed during my internship at Realize, a mobile health support and genotype compatibility system for sickle cell prevention in Cameroon built using UML and the 2TUP process, addresses these challenges by providing families with reliable risk assessments, verified healthcare provider connections, and educational resources tailored to their needs.

III. LIMITATIONS OF THE EXISTING SYSTEM

Table 3 Limitations of the existing system

LIMITATIONS	CONSEQUENCES	PROPOSED SOLUTION
Low public awareness about sickle cell disease and genotype risks	Families ignore testing or underestimate its importance, leading to uninformed family planning decisions	Provide simple educational materials, awareness messages, and preventive guidance accessible on mobile
No centralized system for genotype compatibility checks	Families experience confusion, delays in testing, and increased risk of unplanned sickle cell births	Offer a unified platform with fast, accurate risk results based on genotype input
Lack of clear and accessible information on risk and genotype status	Many people do not know their genotype or compatibility, raising the chances of having affected children	Integrate a user-friendly risk assessment tool and genotype compatibility calculator

Limited availability and visibility of professional healthcare services	People struggle to find help, lose time, or never consult professionals, resulting in preventable complications	Connect users to verified healthcare providers, testing centers, and genetic counselors within the app
Stigma and misinformation in communities	Families may hide their genotype, rely on false beliefs, or delay action	Spread verified knowledge, share real-life guidance, and encourage open, stigma-free discussion
Fragmented and manual process to access services	Families waste time searching or relying on unreliable sources without standardized support	Centralize access by integrating healthcare contacts, provider verification, and educational resources in one app

IV. PROBLEMATIC

In Cameroon, limited awareness about sickle cell disease, delayed genotype testing, and the absence of accessible educational resources contribute to uninformed decisions and high early mortality rates. These issues lead us to the key question: HOW CAN WE PREVENT EARLY MORTALITY FROM SICKLE CELL THROUGH BETTER AWARENESS AND INFORMED DECISIONS IN CAMEROON?

V. PROPOSED SOLUTION

To address the identified challenges, we propose developing a **comprehensive mobile health support and genotype compatibility system** during our internship at Realize. This platform, built using technologies such as MongoDB, React Native, and Node.js, and designed following UML and the 2TUP Process, offers a centralized solution for assessing and managing sickle cell risks. It aims to significantly improve efficiency, transparency, and user experience for families in Cameroon.



- **Accessible Risk Information:** Families can check sickle cell risk easily through genotype compatibility tools and simplified explanations, helping them understand potential outcomes for their children.
- **Awareness and Education:** The app provides clear, easy-to-read resources about sickle cell disease, its risks, and preventive measures, empowering families with knowledge that is often missing in communities.
- **Varied Service Options:** The system provides detailed genotype compatibility analysis, educational resources, and access to genotype testing and genetic counseling, meeting the diverse needs of families from in-depth medical insights to basic guidance.
- **Transparent Information and Reviews:** Verified healthcare providers list their services on the platform, supported by educational content and user feedback to ensure reliability and trust.
- **Trust and Reliability:** Only accredited clinics and trusted medical professionals are highlighted in the app, ensuring families can rely on the information and services they find.

CONCLUSION

The project theme plays a crucial role in defining the scope and identifying the healthcare system for sickle cell risk assessment in Cameroon. Conducting a thorough analysis of the current system during an internship at Realize was essential to understand its limitations, such as lack of organization and limited access to healthcare services. By leveraging information technology and the 2TUP Process alongside UML, we framed the problematic of simplifying sickle cell risk assessment for families and offered a well-formulated solution through Drepalife. This platform tackles the identified issues, empowering parents with accessible, transparent, and reliable tools to predict sickle cell risks for their child, fostering informed healthcare decisions.

FILE 2: SPECIFICATION BOOK

Preamble

The specification book serves as a contractual document before the application development process commences. It succinctly outlines the expectations of the project owner and the respective responsibilities of all parties involved. It provides a specific and detailed description of the topic and user requirements, as well as the necessary conditions for project execution. The project's success will depend on careful allocation and management of the various resources available, including physical, financial, and human assets. Therefore, this document outlines the criteria that the developer must adhere to ensure the success of the project.

Content:

INTRODUCTION

- I. CONTEXT AND JUSTIFICATION
- II. OBJECTIVES OF THE PROJECT
- III. EXPRESSION OF NEEDS
- IV. PROJECT PLANNING
- V. ESTIMATED COST OF THE PROJECT
- VI. LIST OF PARTICIPANTS
- VII. PROJECT CONSTRAINTS
- VIII. DELIVERABLES.

CONCLUSION



INTRODUCTION

The report's specification book enables us to provide comprehensive details about our topic, enhancing our understanding and improving its chances of success. To establish the project's scope, we will describe the context of our topic and identify the issues within it that we aim to solve. Once the solution is proposed, we will outline the objectives we have set for ourselves and for the project. Also, we will assess the functional and non-functional requirements of our system before designing a plan to ensure the timely completion of the project and determine the financial resources needed. Subsequently, we will outline the expected project deliverables at the conclusion of the project.

I. CONTEXT AND JUSTIFICATION

a. Context

Sickle cell disease remains a major public health concern in Cameroon, where limited awareness, fragmented healthcare services, and social stigma continue to hinder effective prevention and management. Many families lack reliable access to information about genotype compatibility, which often results in uninformed marital decisions, delayed diagnosis, and inadequate treatment of affected individuals. As a consequence, the burden of the disease remains high, both medically and socially.

Currently, assessing sickle cell risk and accessing genotype testing or counseling is often informal and inconsistent. Families are forced to rely on incomplete information, traditional beliefs, or irregular access to professional healthcare services. This situation increases the likelihood of preventable complications, poor disease management, and psychological distress for patients and their relatives.

The **DrepaLife** seeks to address these challenges by designing and implementing a digital health support and genotype compatibility platform tailored to the Cameroonian context. The platform will provide accurate risk assessments, educational resources, counseling tools, and verified healthcare provider access through a user-friendly mobile interface. By integrating modern technologies such as secure genotype analysis, real-time data processing, and digital health literacy tools, DrepaLife empowers individuals, couples, and families to make informed decisions and access timely support.

In an era where mobile health solutions are increasingly embraced across Africa, this initiative is both relevant and urgent. DrepaLife positions itself as an innovative response to improve awareness, strengthen prevention, and ultimately reduce the incidence and impact of sickle cell disease in Cameroon.

b. Justification

The primary justification for this project lies in the persistent gaps in sickle cell awareness, prevention, and healthcare access in Cameroon. A significant number of families do not have access to reliable genotype testing, professional counseling, or clear and structured information about sickle cell disease. Existing resources are fragmented, often informal, and



lack transparency, which leaves families without the guidance needed to make informed marital and health decisions. This situation leads to delayed genotype testing, widespread misinformation, poor disease management, and preventable complications or early deaths among patients.

The **DrepaLife** addresses these challenges by creating a simple, accessible, and trustworthy digital health platform that:

- **Improves Access:** Enables families to check genotype compatibility risks and connect with nearby verified healthcare providers through an intuitive mobile interface.
- **Raises Awareness:** Delivers clear and culturally adapted educational resources on genotype compatibility, sickle cell prevention strategies, and practical daily health guidance.
- **Supports Families' Needs:** Offers flexible tools ranging from quick compatibility checks to comprehensive educational content, tailored to users with varying levels of health literacy.
- **Builds Trust:** Features only verified healthcare providers and accredited information sources, ensuring credibility and user confidence.

By aligning with the rapid growth of mobile health solutions in Africa, DrepaLife empowers families with the knowledge, resources, and tools needed to prevent new sickle cell cases, encourage early diagnosis, and significantly improve long-term disease management in Cameroon.

II. OBJECTIVES OF THE PROJECT

A. GENERAL OBJECTIVE

The main objective is to improve healthcare access and management for sickle cell patients by connecting them with qualified medical professionals, providing reliable health tips, and offering essential tools such as crisis management guides and genotype compatibility checks

B. SPECIFIC OBJECTIVES

- Facilitate Easy Access to Medical Information
- Enhance User Experience
- Ensure Transparency in Medical Consultations.
- Enable Health Record Management
- Ensure Timely Medical Assistance
- Improve Accuracy of Genotype Compatibility Checks

The project is characterized by the following:

- **Project Name:** Drepalife
- **Project Target:** Individuals living with sickle cell disease, families seeking genotype compatibility information, and health professionals managing sickle cell patients.
- **Technical Specification:** Mobile application (React Native) integrated with a backend (Node.js, MongoDB) for real-time data management, secure authentication, genotype compatibility analysis, and online payment processing for consultations.

III. EXPRESSION OF NEEDS

1. Functional Needs

In this context, functional requirements describe what the system or application should do. The modules are as follows;

Patient:

- **Create Account:** Patients can register and fill in personal and medical information relevant to sickle cell management (e.g., genotype).

- **Submit Daily Health Status:** Patients regularly submit updates on their current health condition.
- **Perform Crisis Self-Help Procedure:** Patients can follow guided steps for managing a sickle cell crisis on their own.
- **Request Genotype Compatibility Test:** Patients can request an analysis of genotype compatibility for family planning.
- **Request Expert Consultation:** Patients can request medical consultations with health experts through the app.
- **Make Secure Payment for Consultation:** Patients make secure payments for premium consultation services through integrated payment APIs.
- **Chat with an AI chatbot:** Patient can chat with a well-trained medical chatbot
- **Rate Health Expert:** Patients provide feedback and rate the health expert after consultations.

Health Expert / Medical Doctor:

- **Set Profile:** Health experts create and update their professional profiles, including qualifications and specialties.
- **Respond to Consultation Requests:** Experts review and respond to patient consultation requests.
- **Provide Recommendations:** Health experts offer personalized health advice and crisis management guidance.
- **Track Patient's Health Progress:** Experts monitor and track ongoing health data submitted by patients.
- **Update Crisis Management Protocol:** Experts update and validate protocols and educational content related to crisis care.
- **Publish and Update Health Tips:** Admins manage the publication of daily health tips and educational content.

Administrator:

- **Register and Approve Health Experts:** Admins verify and approve health expert accounts and credentials.
- **Manage User Accounts:** Admins oversee the registration and management of all users, including patients and experts.
- **Validate Crisis Management Content:** Admins review and approve updated crisis management protocols and health tips.

2. Non-Functional Needs

In this section, non-functional requirements define the quality attributes, standards, and performance measures that the system must adhere to. These aspects ensure the overall performance, security, and usability of the application.

1. Performance Requirements

- **Handle Concurrent User Actions:** The system must support multiple patients submitting health statuses, requesting consultations, or performing genotype compatibility tests simultaneously without lag, especially during peak usage times.
- **Fast Response Times:** Actions such as submitting health status, receiving compatibility results, or initiating a consultation must be completed in under 3 seconds for a seamless user experience.

2. Security Requirements

- **Data Encryption:** Encrypt all sensitive data, including health profiles, payment details, and genotype data, to protect user actions like submitting health status or requesting genotype tests.

3. Scalability Requirements

- **Support Growing User Actions:** The system must scale to handle increasing numbers of patients submitting daily health statuses, requesting consultations, or performing genotype tests without performance degradation.
- **Horizontal Scaling:** Enable horizontal scaling for both frontend (e.g., patient app interface) and backend (e.g., genotype analysis service, consultation processing) to support growing demand for actions like health status updates and expert responses.

IV. PROJECT PLANNING

a. Chronogram of activities

We began our internship on the 1st of July, till the 30th of September 2025, a total of 3 months. To this, it was an obligation for us to manage our time properly so as to finish on time. This section presents how the work phases were scheduled throughout my internship period.

Table 4 Chronogram of activities

PHASE	OBJECTIVE	OUTPUT	DURATION	PERIOD
INSERTION	Collection of information on the enterprise	Insertion book	2 weeks	08 th July to 19 th July
EXISTING SYSTEM	Study of the existing system	Existing System	1 week	22 nd July to 26 th July
SPECIFICATION BOOK	Specification of the user needs	Specification Book	1 week	29 th July to 02 nd August
ANALYSIS	Capture of needs Use case and textual description Modelling	Analysis Book	2 weeks	05 th August to 16 th August
CONCEPTION	Preliminary conception and Detailed conception	Conception book	2 weeks	19 th August to 30 th August
REALIZATION	Implementation Unitary test Integration Test Development	Realization book	3 weeks	02 nd September to 20 th September

TEST OF FUNCTIONALITIES	Testing of the software and debugging	Test of functionalities	1 week	23 rd September to 27 th September
INSTALLATION AND USER GUIDE	Documenting software	User Guide	1 week	30 th September to 04 th October

b. Gantt Project planning



Figure 3 Drepalife gantt project planning

V. ESTIMATED COST OF THE PROJECT

a. Software Resources

For the realization of this project, we will need the following software applications:

Table 5 Software resources for Drepalife (Source Mercurial 2024)

RESOURCES	DESIGNATION	USAGE	QUANTITY	UNIT COST (FCFA)
Formatting	MICROSOFT OFFICE PROFESSIONAL 2019	Writing of the report	1	355465
Mobile Testing	Expo Go (Mobile Device)	Testing React Native-based mobile app features	1	Freeware
Code Editor	Visual Studio Code	For writing the code	1	Freemium

		of the application		
Project Planning	Gantt Project	For building a Gantt chart	1	Freemium
UML Analysis	Visual Paradigm	Drawing UML diagrams to map action	1	Community Version
Version Control	Git & GitHub	For tracking and managing changes to software code	1	Free Version
TOTAL 1			6	355465

b. Hardware Resources

This includes the material resources used to realize this project. The table below shows a representation of these different materials:

Table 6 Hardware Resources for drepalife (source: <https://www.scribd.com/document/561202966/mercuriale-2024>)

RESOURCES	HARDWARE	QUANTITY	UNIT COST (FCFA)
Computer	Lenovo Desktop-Q1U5165 (Intel Core i7-5600U, 8GB RAM, 256GB SSD)	1	100,000
Network	Local network installation	1	226,000
Removable Disk	USB HP 2.2 FLASH DRIVE V225W - 8 GB	1	4,160
Smartphone	Google Pixel 6 (256GB)	1	110,000
Total 2		4	440,160

c. Human resources

Table 7 Human resource for drepalife (Source Mercurial 2024)

ROLE	NUMBER OF DAYS	QUANTITY	COST PER DAY	TOTAL PRICE (FCFA)
Project Manager	90	01	30 000	2 700 000
Analyst	21	01	25 000	525 000
UI/UX Designer	30	01	10 000	140 000
Programmer	30	01	15 000	450 000
Tester	14	02	10 000	140 000
TOTAL 3				3 955 000

d. Global Estimation

Table 8 Global estimation

TOTAL 1 (FCFA)	TOTAL 2 (FCFA)	TOTAL 3 (FCFA)	Hosting Cost (Vercel)	Unexpected charges (10%)	OVERALL TOTAL (FCFA)
355465	440,160	3,955,000	133,989	475,063	5,359,677
Five million, three hundred fifty-nine thousand, six hundred seventy-seven					

VI. LIST OF PARTICIPANTS

The table below presents the different persons who took part in the accomplishment of this project. They include:

Table 9 list of participants

NAME	FUNCTION	ROLE
Mr. TSEMBOM PERCY	Follows up with interns at the company level	Professional Supervisor
Mrs. NZALI CLAUDIA	Follows up with the student at the academic level	Academic Supervisor
Mrs. NGOUETSO SANDRA	Follows up with interns at the company level	Software Engineer
Mrs. CHAYIE LUCRESE	Follows up with interns at the company level	Software Engineer
Mr. MENGOT NSO	Follows up with interns at the company level	Software Engineer
METOMO JOSEPH EMMANUEL RALPH	AICS Student Intern	Student at AICS

VII. PROJECT CONSTRAINTS

a) Software constraints

- ❖ **Compatibility:** The platform must be compatible with different devices (smartphones, tablets) to reach a wide audience.
- ❖ **Security:** Robust security measures must be put in place to protect users' personal data and prevent any breach of confidentiality.
- ❖ **User-friendliness:** The user interface must be user-friendly to ensure a positive user experience.

b) Material constraints

- ❖ **Internet availability:** The platform assumes that users have access to the Internet. However, efforts must be made to guarantee an acceptable experience even with limited Internet connections.
- ❖ **Hardware compatibility:** The platform must be designed to run on a variety of devices, taking into account differences in screen size and processing power.

c) Quality constraints

- ❖ **Accuracy of information:** Guides and information provided must be accurate, up-to date and verified, to avoid giving incorrect information to users.
- ❖ **Performance:** The platform must be fast and responsive to avoid excessive loading times and maintain user engagement.
- ❖ **Accessibility:** The platform must be designed to be accessible to people with disabilities, in accordance with web accessibility standards.

d) Time constraints

- ❖ **Deadlines:** The project must be completed within the specified deadlines to meet user needs and stakeholder expectations.
- ❖ **Regular updates:** Future updates and enhancements to the platform must be planned to meet the changing needs of citizens and to maintain long-term relevance.



VIII. DELIVERABLES

In this section, we will present the different elements that will be delivered at the end of this project. They include:

- Installation and user guide
- Software Application
- PowerPoint presentation



CONCLUSION

To conclude, the specification book has permitted us to present the general and specific objectives of the project, the project constraints, the functional and nonfunctional needs of the project, lay out the different resources of the project, and finally, the project planning. The end of this document will enable us to easily dive into drafting the analysis document, which will bring out the technical aspects of the project as a whole.



FILE 3 : THE ANALYSIS PHASE

Preamble

The analysis document that adheres to the specifications provides a comprehensive overview of the issue and the proposed solution, supported by diagrams. It outlines the selected analysis method for the project and explains the rationale behind this choice. All of these elements contribute to a clearer understanding of the information system being modeled.

Content:

INTRODUCTION

I. METHODOLOGY

- a. COMPARATIVE STUDY BETWEEN MERISE AND UML
- b. PRESENTATION OF THE 2TUP
- c. CHARACTERISTICS OF THE 2TUP
- d. JUSTIFICATION OF THE CHOICE OF THE ANALYSIS

II. MODELING

- a. USE CASE DIAGRAM
- b. COMMUNICATION DIAGRAM
- c. SEQUENCE DIAGRAM
- d. ACTIVITY DIAGRAM

CONCLUSION

INTRODUCTION

System development can be thought of as having two major components: System analysis and system design, which both help in understanding the details of the existing system or the system to be designed. The analysis and design of information systems has most of the time vocation to allow the creation of databases, which must represent as closely as possible the reality of the field studied thus requiring the use of a design method. This is why our choice will be directed on the UML method as it offers much to developers seeking a user-centered approach and / or a wide scope in design. This part of the report consists of the comparative study of UML and MERISE, unified processes and finally the various diagrams that meet the functional need requirements.

I. METHODOLOGY

Analysis is the fundamental step in the conception of software. It is the basis for the realization of every information system. An information system is an organization of resources (human, software, and hardware), data, and feedback to attain a specific objective. Several methods and languages have been put in place to facilitate the analysis and conception of information systems, which principally have UML and MERISE.

a. COMPARATIVE ANALYSIS BETWEEN MERISE AND UML

i. MERISE

MERISE stands for “Méthode d’Etude et de Réalisation Informatique pour des Systèmes d’Entreprise”. Although it is prescriptive to some extent, MERISE permits the participation of end users and senior management as well as data processing professionals in its decision cycle. MERISE is a method for designing, developing and carrying out IT projects. The goal of this method is to achieve the design of an information system. The MERISE method is based on the separation of data and processing to be carried out in several conceptual and physical models. The essentials of the approach lie in its three cycles: the decision cycle, the life cycle and the abstraction cycle, which cover data and process elements equally. The separation of data and processing ensures longevity in model. Indeed, the arrangement of data does not have to be often overhauled, while treatments are more frequently.

ii. UML

UML (Unified Modelling Language) is a standard notation for the modelling of real-world objects as a first step in developing an object-oriented design methodology. Its notation is derived from and unifies the notations of three object-oriented design and analysis methodologies: Grady Booch's methodology for describing a set of objects and their relationships, James Rumbaugh's Object-Modelling Technique (OMT), Ivar Jacobson's approach which includes a use case methodology. Other ideas also contributed to UML, which was the result of a work effort by Booch, Rumbaugh, Jacobson, and others to combine their ideas, working under the sponsorship of Rational Software. UML captures information about the static and dynamic view of a system. UML 2.5 comprises of 14 diagrams which represent the different views of a system. The 14 diagrams can be subdivided into two, Static or structural and Dynamic diagrams. These diagrams include;

1. STATIC OR STRUCTURAL DIAGRAMS

- Class diagram;
- Object diagram;
- Component diagram;
- Deployment diagram;
- Composite Structure diagram;
- Package diagram;
- Profile Diagram;

2. BEHAVIOURAL OR DYNAMIC DIAGRAMS

- Use case diagrams;
- Activity diagram;
- State machine diagram;
- Sequence diagram;
- Communication diagram;
- Global Interaction diagram;
- Timing Diagram;

These diagrams can be represented on the diagram bellow:

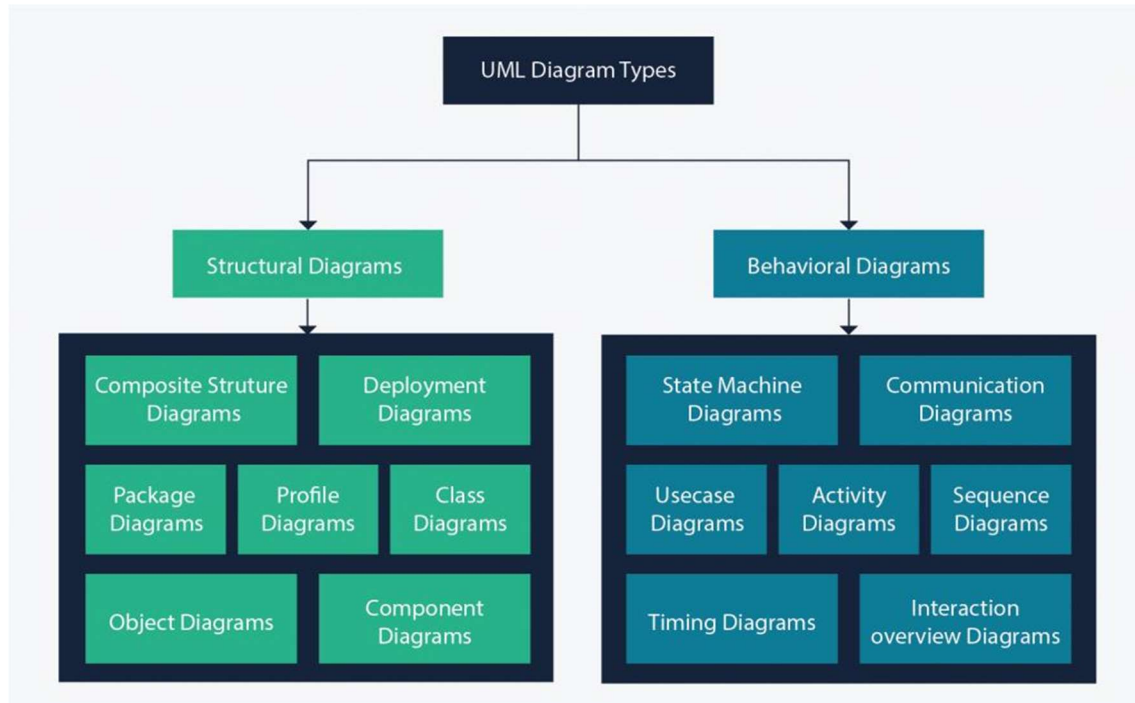


Figure 4 UML Structural and behavioural diagram

Source: <https://images.edrawsoft.com/articles/what-is-uml/uml-diagram-types.png>

Table 10 Comparative analysis between MERISE and UML

MERISE	UML
Méthode d'Etude et de Realization Informatique pour les Systèmes d'Entreprise	Unified Modeling Language
Merise is a method of analysis.	UML is a modeling language
Merise is a systematic method of analysis and design for information systems, i.e., it uses a system approach.	UML is, however, not a method but rather an object-oriented language that uses the 2TUP approach to model a system
It is based on a procedural approach	Based on the object approach
Merise proposes to consider the real system from two points of view: • The Static view(data) • The Dynamic view(treatments). Which means in Merise, we have a separate study of data and treatments	UML offers a different approach from that of MERISE in that it combines data and process

b. PRESENTATION OF THE 2TUP

2-TUP (Two Track Unified Process) is Y in shape and is a unified process that has as its objective to bring a solution to constraints of functional changes imposed on information systems. It proposes a development cycle that separates the functional aspects from the technical aspects. 2-TUP therefore distinguishes two main branches:

(The functional and technical branch) whose different results are being merged to realize our system.

Any unified process UP (Unified Process) process meets the following characteristics:

- ❖ It is iterative and incremental.
- Some of the advantages of the 2TUP include:
- ❖ It allows the addition of modules.
- ❖ It allows flexibility in the analysis of a project.
- ❖ It is easy to design and understand.
- ❖ It allows to have a preliminary view

Below is the representation of 2-TUP:

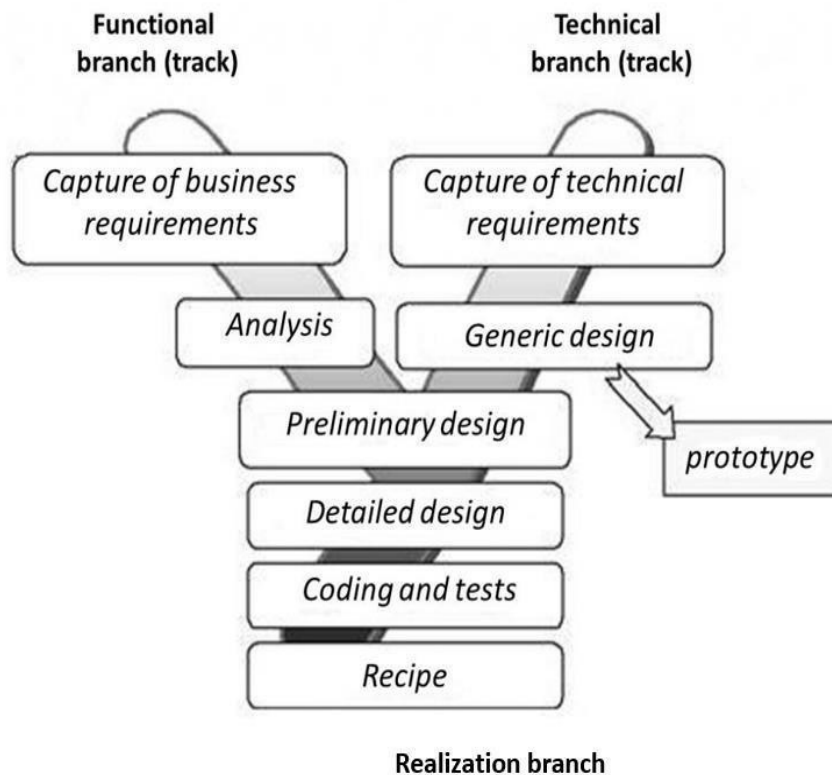


Figure 5 2TUP diagram(Source <https://www.allacronyms.com/2TUP/2TrackUnifiedProcess>)

i.) The Functional Branch:

This branch captures business needs, resulting in a model focused on the end-user's requirements.

ii.) The Technical Branch:

This branch proposes a generic design from the technical needs, identifies the materials, tools, and technologies to be used.

iii.) The Implementation:

The implementation phase consists of bringing the two branches together, providing the application design, and finally the delivery of a solution adapted to the needs.

a) Functional Branch (left branch)

- ❖ Capture of Business Needs: This part minimizes the risk of producing an inadequate system with the needs of users and also verifies its consistency.
- ❖ Analysis: This is the study of the specifications to find out what the system will be made of in terms of the business rules. We have the following diagrams: Use Case, Activity Diagram, State Machine Diagram, and Communication Diagram.



b) Technical Branch

- ❖ **Capture of Technical Needs:** This stage consists of the identification of tools, materials, and technologies to use in developing the system. This technical architecture will be presented in this stage.
- ❖ **Generic Designs:** The technical architecture will be presented in this stage. We have the following diagram: mockUp7 Design.

c) Implementation Branch (middle branch):

- ❖ **Preliminary Branch:** This is the stage where the analysis model is integrated into the technical architecture. The goal here is to know what technical component will be used depending on the features of the analysis. We have the following diagrams: interaction overview, Component, Deployment, package diagram and Composite Diagram.
- ❖ **Detailed Design:** this is the detailed design of each feature of the system. We have the following diagrams: class, object, sequence and timing Diagrams.
- ❖ **Coding and Testing:** This is the programming phase of the designed features, alongside testing of the coded features.
- ❖ **Recipe (Result):** This is the validation phase of the functions of the system developed.

c. CHARACTERISTICS OF 2TUP

The two-track unified process was built to be used with the Unified Modeling Language to become a method and has the following characteristics:

- ❖ **It is user oriented:** 2TUP is built from user's expectation
- ❖ **It is component-oriented:** it offers flexibility to the model by supporting the re-use of components.
- ❖ **It is an iterative process:** processes are being done in iterations, and each iteration shows a precise level of abstraction.
- ❖ **It is an incremental process:** allowing a better functional and technical risk management and thus constituting the deadlines and the cost control.

d. JUSTIFICATION OF THE CHOICE OF THE ANALYSIS

Our choice was based on the modeling language UML, associated with it 2TUP, due to certain criteria, which are as follows:

- ❖ UML is based on an object-oriented approach;
- ❖ UML associate's data and treats it;
- ❖ It is a modeling language that focuses mainly on users' needs and expectations;
- ❖ 2TUP is a process built from an object-oriented approach and is built from UML.
- ❖ 2TUP facilitates the modeling of complex systems, which can further be readjusted.

II. MODELLING

a. USECASE DIAGRAM

i. DEFINITION

The use case diagram simply shows the relationship between the actors of the system and the system. It permits users to easily visualize the different actions that can be done by the users. It also clearly defines the boundaries of the system.

ii. FORMALISM

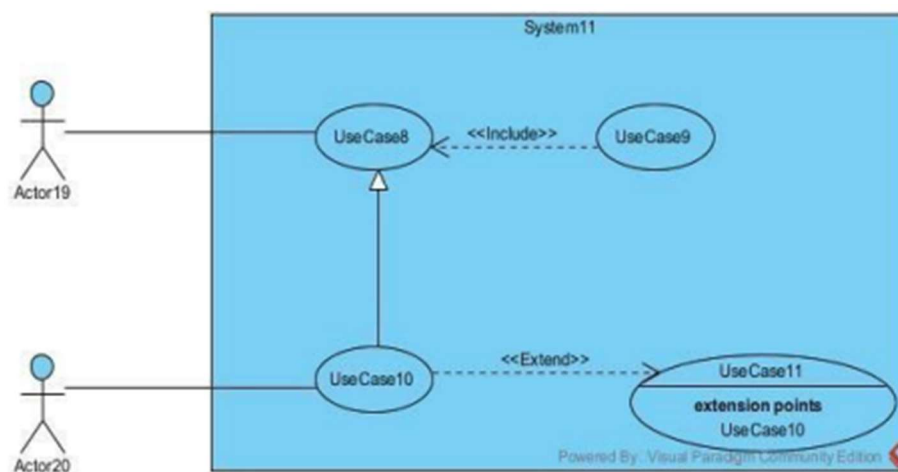
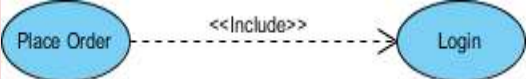
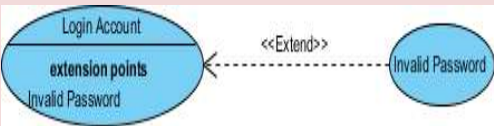
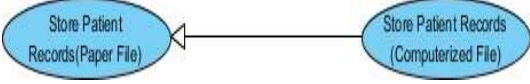


Figure 6: Formalism of use case diagram

iii. COMPONENTS OF USE CASE DIAGRAM

Table 11 Component of use case diagram

Element	Notation	Description
Actors		Represents an entity that directly interacts with the system. The actor is what performs the different possible actions of the system
Use case		A use case represents a functionality of the system. It is an action that can be performed by an actor.
Association		it indicates that an actor takes part in a use Case
Include		An inclusion denotes that an included action must be performed before the including action can be performed.
Extend		An extension denotes that an extending action may be performed while an extended action is being performed.

Generalization		This shows that an actor or a use case is a kind of another abstract or concrete actors which can be defined and later specialized using generalization relationship.
System		It is a container of use cases which interact with external actors

iv. ACTORS OF OUR SYSTEM

➤ Primary Actors

Table 12 Actors of our system

Actor	Role
Patient	The primary user of the app, a patient actively manages their health by submitting daily health statuses, requesting expert consultations, performing crisis self-help procedures, and making secure payments for consultations via the Payment API. They also receive automated reminders for medication, hydration, or appointments and provide feedback on consultations.
Health Expert	Health experts deliver medical support by reviewing patient health status submissions, responding to consultation requests via chat or video calls, providing personalized health recommendations, and updating crisis management protocols. They manage their profiles to ensure accurate credentials and availability for patient interactions.

Administrator	Admins ensure platform efficiency by registering and approving health experts, managing patient accounts (e.g., activating or deactivating), publishing and validating health tips, monitoring consultation records, tracking payments processed through the Payment API, and generating platform usage reports.
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➤ Secondary actors

Table 13 Secondary Actors of drepadata

Actor	Role
Payment API	This external system processes financial transactions by validating payment information, executing secure payments for patient consultations, and sending confirmation receipts to the app, enabling seamless and secure consultation bookings.
Chatbot API	This external system provides an interactive, conversational assistant within the app. It helps families quickly access information about sickle cell disease, genotype compatibility, and preventive measures without needing to search through multiple sections.

v. GENERAL USE CASE DIAGRAM

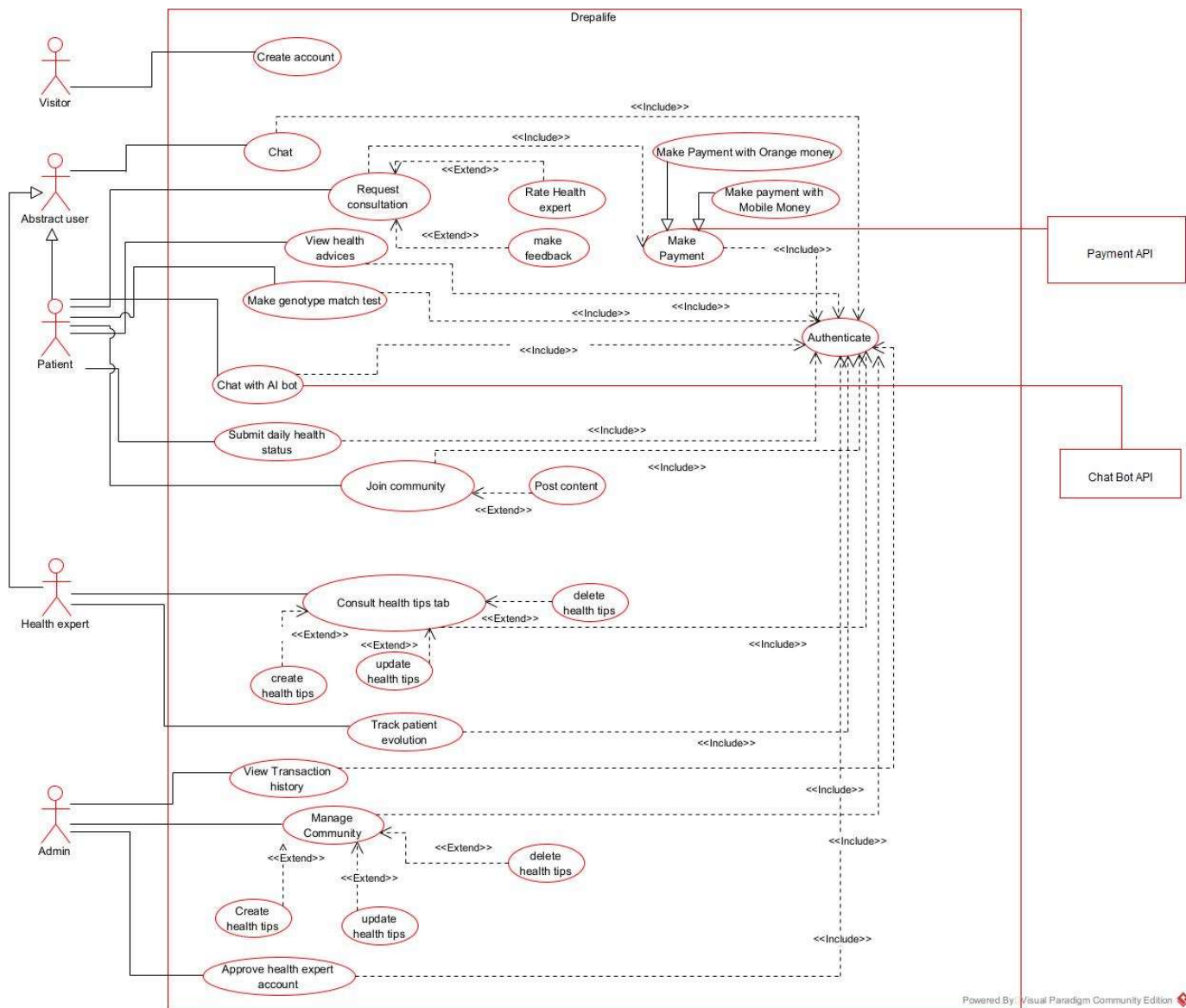


Figure 7 General Usecase of Drepalife

vi. SPECIFIC USE CASE DIAGRAM FOR MAKE GENOTYPE MATCH TEST

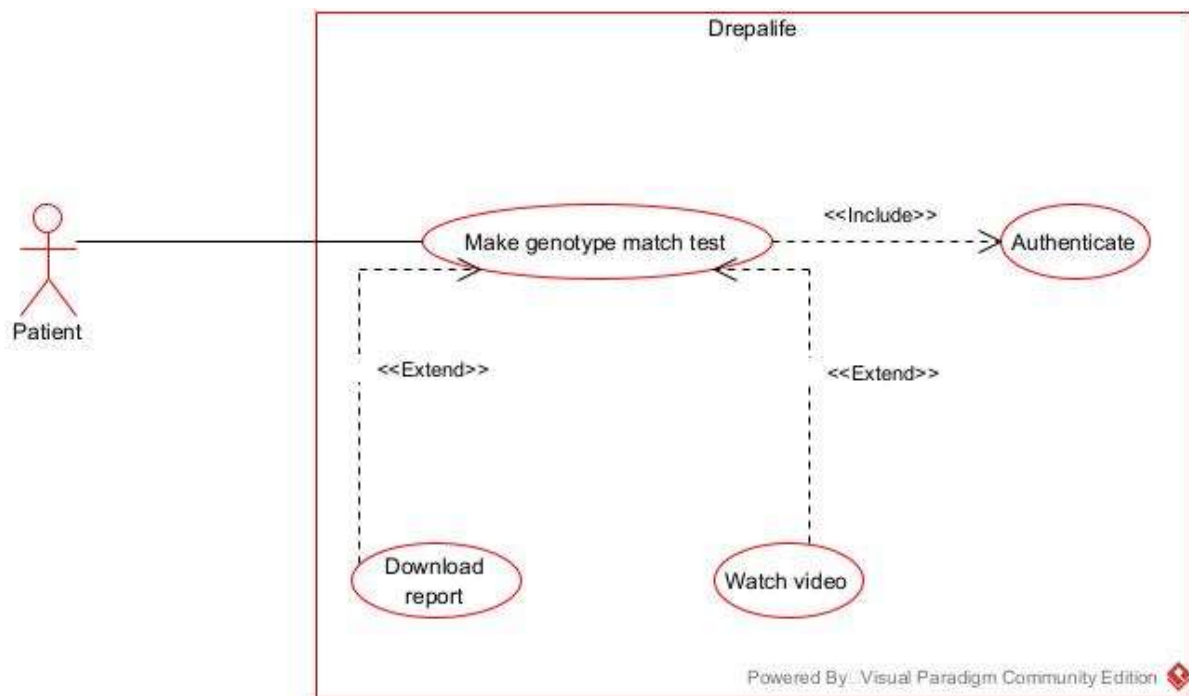


Figure 8 specific diagram for make genotype compatibility match test

vii. TEXTUAL DESCRIPTION OF USE CASES

➤ Authenticate

Table 14 textual description of usecase

USECASE AUTHENTICATE	
Summary	Permit Patient to access the system
Actors	Patient, Health Expert
Pre-condition(s)	1. The actor already has an account; 2. The app is launched 3. The account must be active
Trigger	The user clicks on login button
Nominal scenario	1. The user clicks on login button 2. The system displays the login page 3. The user fills form and submit 4. The system checks conformity 5. The system sends validation query 6. The system treats the result returned by the database 7. The user is directed to his page
Alternative scenario	8. If step 4 of the nominal scenario the format is not correct, an error message is returned to the user and he returns to step 3 of the nominal scenario. 9. If step 6 of the nominal scenario no user was found, the system returns an error message and returns to step 3 of the nominal scenario
Post-condition	The user logs in successfully
of Success	
Post condition of failure	An error message is display and user does not have access his space.

➤ Make payment

Table 15 Textual description of make payment

Actions	Patient
Summary	Permit patients to pay for requesting a consultation
Actors	Patient, Payment API
Pre-condition(s)	1. The app is launched 2. The patient has successfully requested a consultation 3. The Health expert has confirmed the Request
Trigger	The user clicks on payment button
Nominal scenario	1. The user clicks on payment button 2. The system displays payment form 3. The patient fills form and submit 4. The system checks conformity 5. The system sends payment request 6. The system treats the result returned by the payment API 7. The system sends payment information to the database 8. The system treats the result returned by the database 9. The system displays a successful message
Alternative scenario	10. If step 6 of the nominal scenario the format is not correct, an error message is returned to the patient and he returns to step 2 of the nominal scenario.

	13. If step 8 of the nominal scenario an error is found, an error message is returned to the patient and he returns to step 2 of the nominal scenario 14. If step 10 of the nominal scenario an error is found, an error message is returned to the patient and he returns to step 2 of the nominal scenario
Post-condition of success	If successful, the patient receives a successful message. The system keeps a record of the payment transaction.
Post condition of failure	An error message is displayed

➤ **Make Genotype Compatibility Match Test**

Table 16 Textual description make genotype compatibility match test

Actions	Patient
Summary	Permit patients to perform a genotype compatibility test by submitting genotype information and receiving test results.
Actors	Patient
Pre-condition(s)	1. The app is launched 2. The patient has been authenticated (logged in) 3. The patient has access to genotype test functionality
Trigger	4. The patient clicks on the “Genotype Test” button

Nominal scenario	5. The patient clicks on the “Genotype Test” button 6. The system displays the genotype test form 7. The patient fills in and submits the genotype test form 8. The system checks if the submitted data is valid 9. If valid, the system displays a success message 10. The system sends a genotype fetch query to the DBMS 11. The DBMS executes the query and sends a response 12. The system verifies the response 13. If the response is correct, the system displays the genotype test result
Alternative scenario	14. If in step 4 the data is invalid, the system displays an error message and returns to step 2. 15. If in step 8 the response from the DBMS is invalid, the system displays an error message and stops the process.

b. COMMUNICATION DIAGRAM

i. DEFINITION

The communication diagram is used to model the specific interaction between the participants of the system. It describes both the static and dynamic behavior of the system. It shows the chronological succession of the actions performed by an actor to realize an operation.

ii. FORMALISM

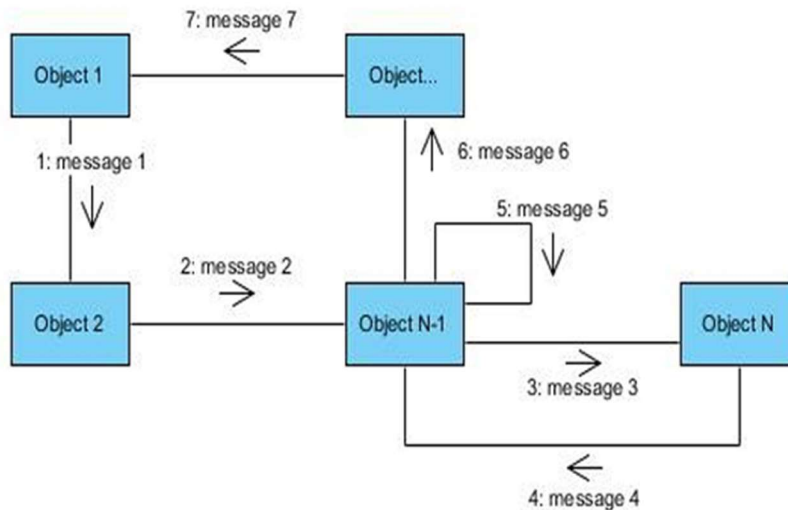
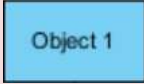


Figure 9 Formalism of communication diagram

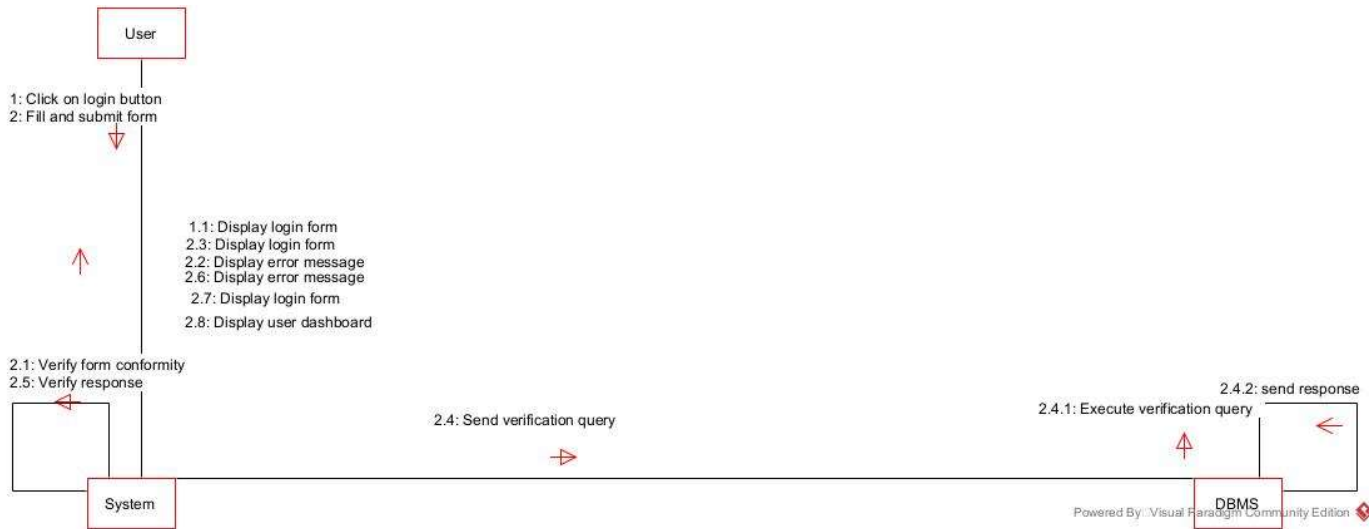
iii. COMPONENTS OF COMMUNICATION DIAGRAM

Table 17 Components of communication diagram

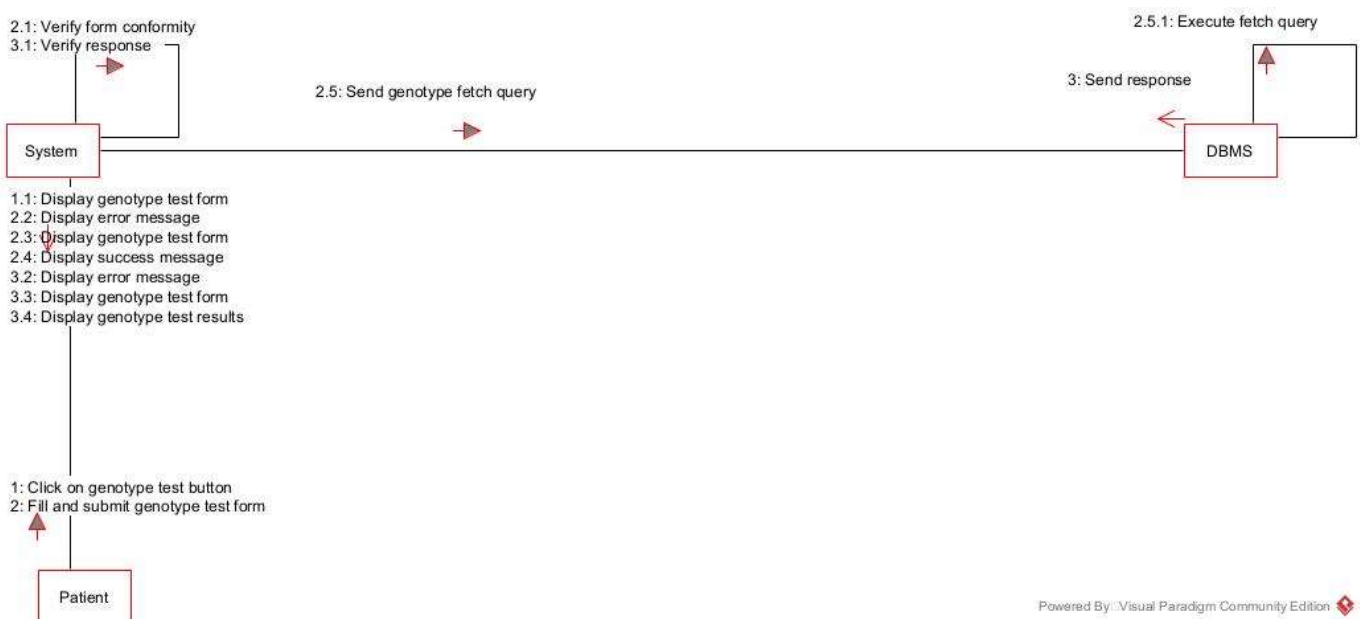
Element	Notation	Description
Message	2: message 2 →	Designs a particular communication between lifelines.
Actors		A role play by an entity that interacts with the subjects
Link		It initiates an association it connects two objects together for them to communicate.

Object		An actor represents an individual participant in the interaction conversation
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iv. AUTHENTICATE COMMUNICATION DIAGRAM



v. MAKE GENOTYPE TEST MATCH COMMUNICATION DIAGRAM



vi. MAKE PAYMENT COMMUNICATION DIAGRAM



Figure 12 make payment communication diagram

c. SEQUENCE DIAGRAM

i. DEFINITION

A sequence diagram is a UML diagram that illustrates the sequence of messages between objects in an interaction. It consists of a group of objects that are presented by lifelines and the messages that they exchange over time during the interaction. These interactions are arranged from top to bottom following their order of occurrence.

ii. FORMALISM

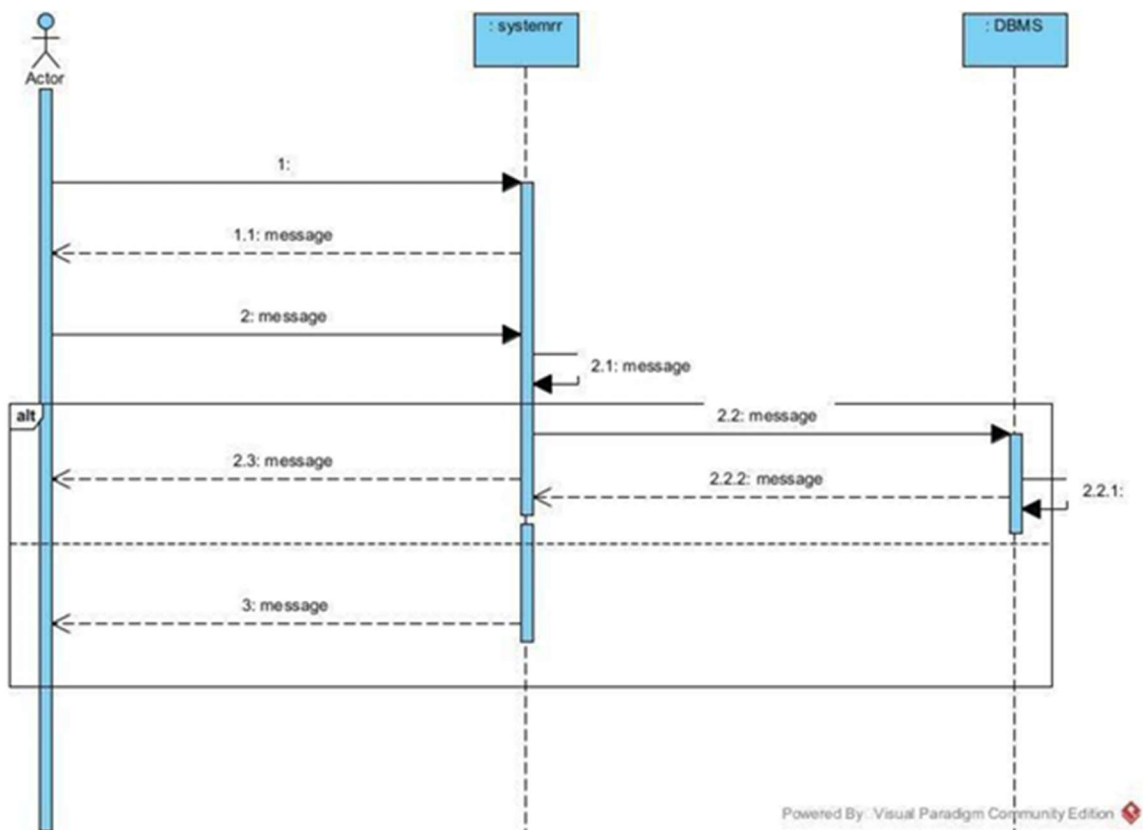
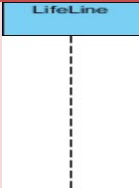
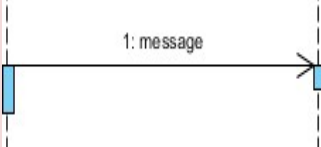
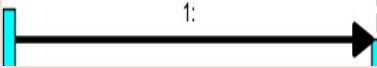
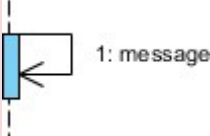
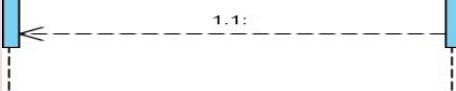


Figure 13 Formalism of sequence diagram

iii. COMPONENTS OF SEQUENCE DIAGRAM

Table 18 Components of sequence diagram

Element	Notation	Description
Lifelines		They represent rows or objects instances that participate in the sequence being modelled.
Asynchronous Message		It is a message that receives an indirect response.
Synchronous Message		It is a message that sends and want response before it continues a process
Self-Message		Self message is a kind of message that represents the invocation of message of the same lifeline.
Return Message		It represents the response of a message.
Actor		They send and receive message.
Combined Interaction on Fragment		An articulation of interaction diagram, defined by an operator and operands.
Object		They send and receive messages

iv. AUTHENTICATE SEQUENCE DIAGRAM

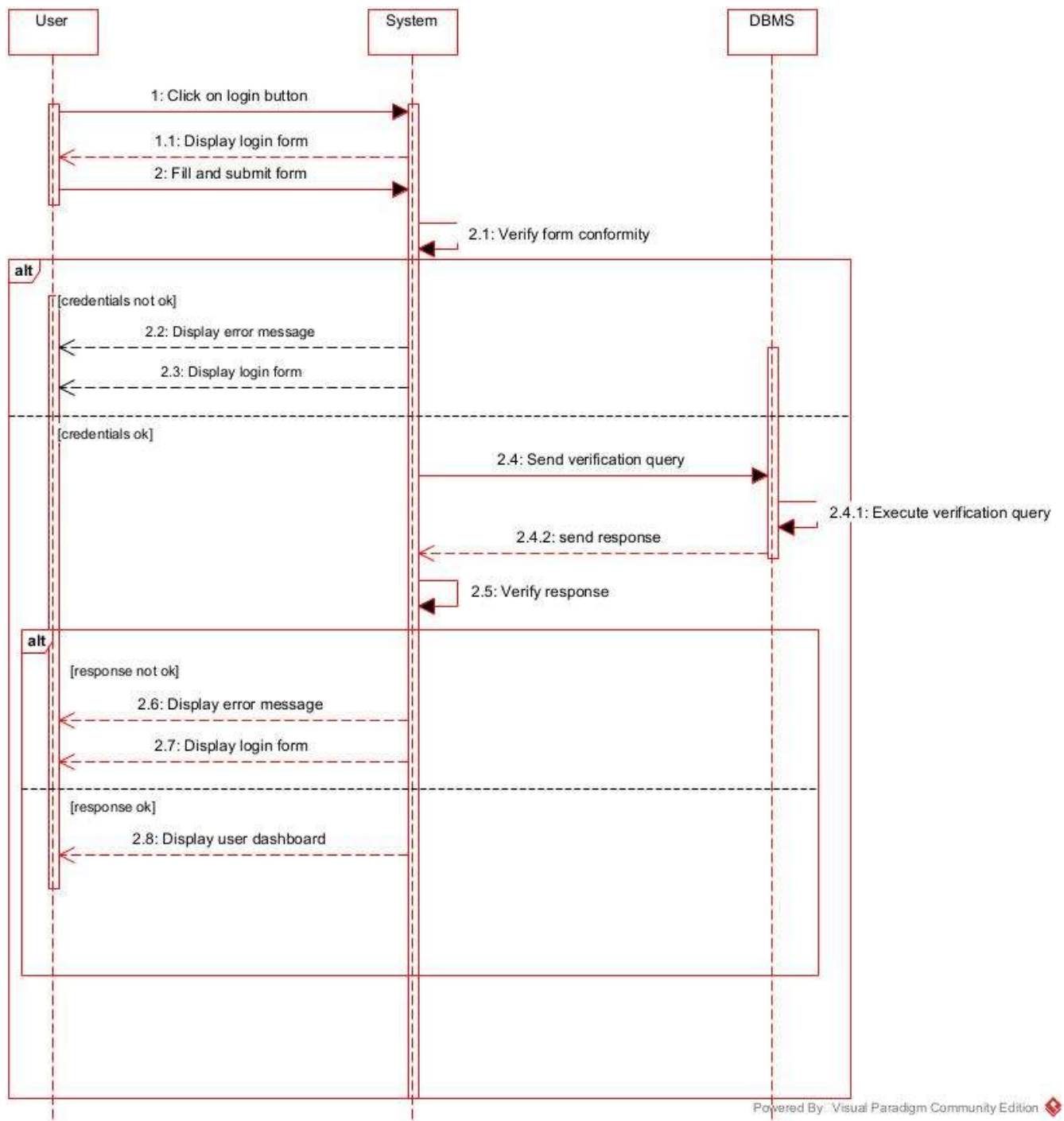


Figure 14 Authentication sequence diagram

v. MAKE PAYMENT SEQUENCE DIAGRAM

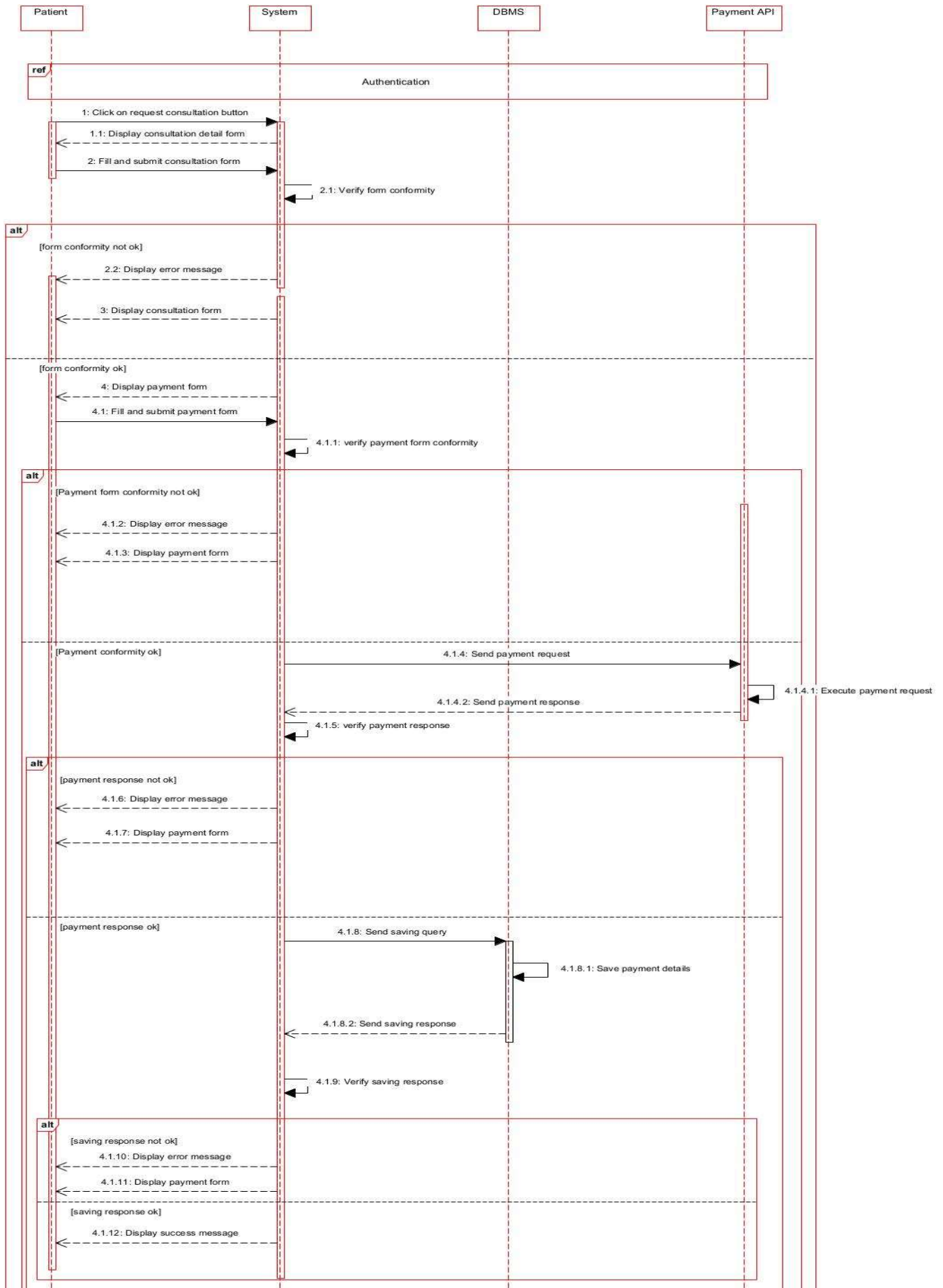


Figure 15 make payment sequence diagram

vi. MAKE GENOTYPE TEST MATCH SEQUENCE DIAGRAM

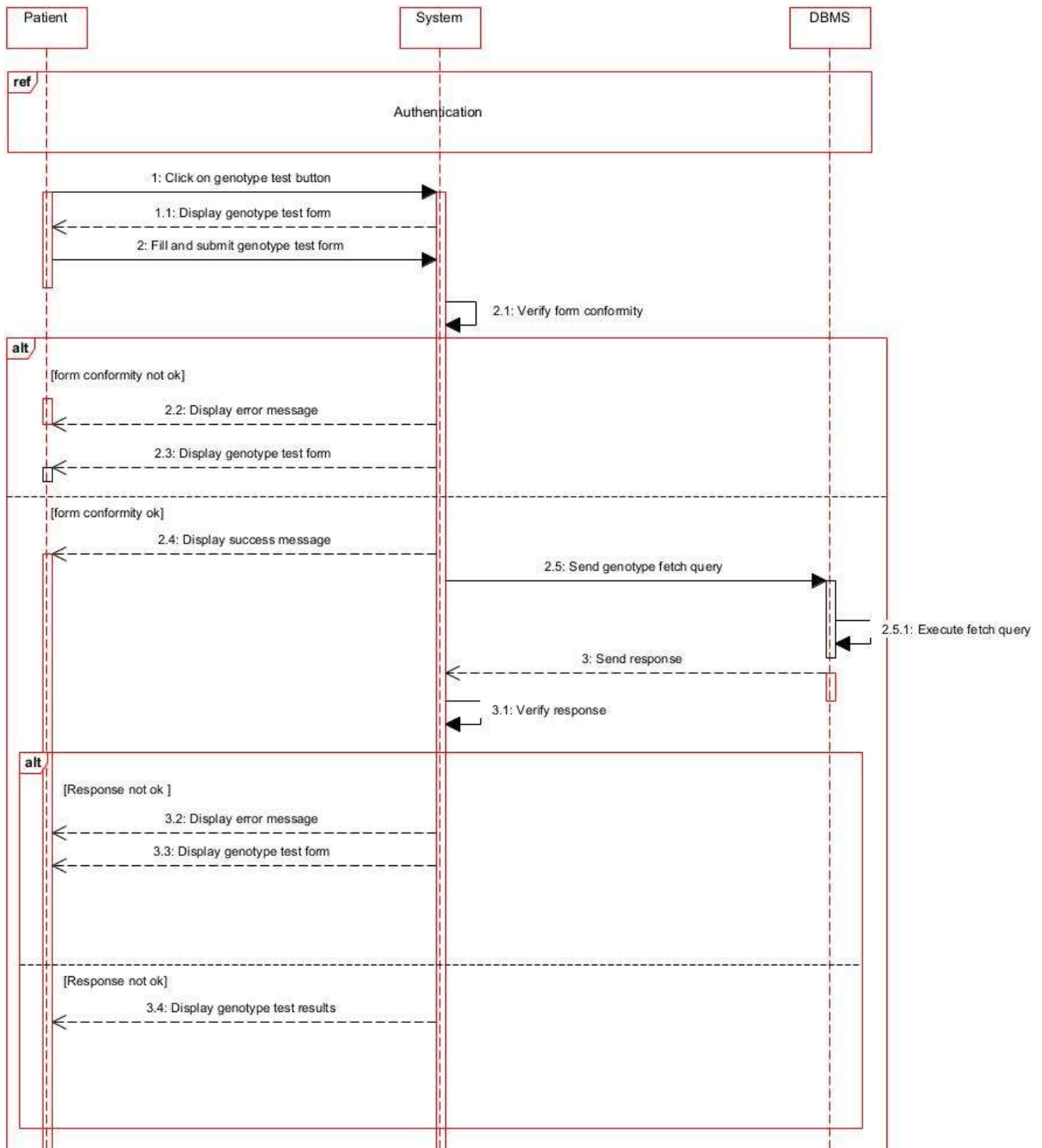


Figure 16 make genotype test match sequence diagram

d. ACTIVITY DIAGRAM

i. DEFINITION

An activity diagram is a graphical representation of workflows that show the steps needed in the realization of a process, showing the details from a start point to an end point through all decisions and actions that can possibly be performed. Activity diagrams are intended to model both the computational and organizational processes. The flow can be sequential, branched, or concurrent. Below is an activity diagram formalism.

ii. FORMALISM

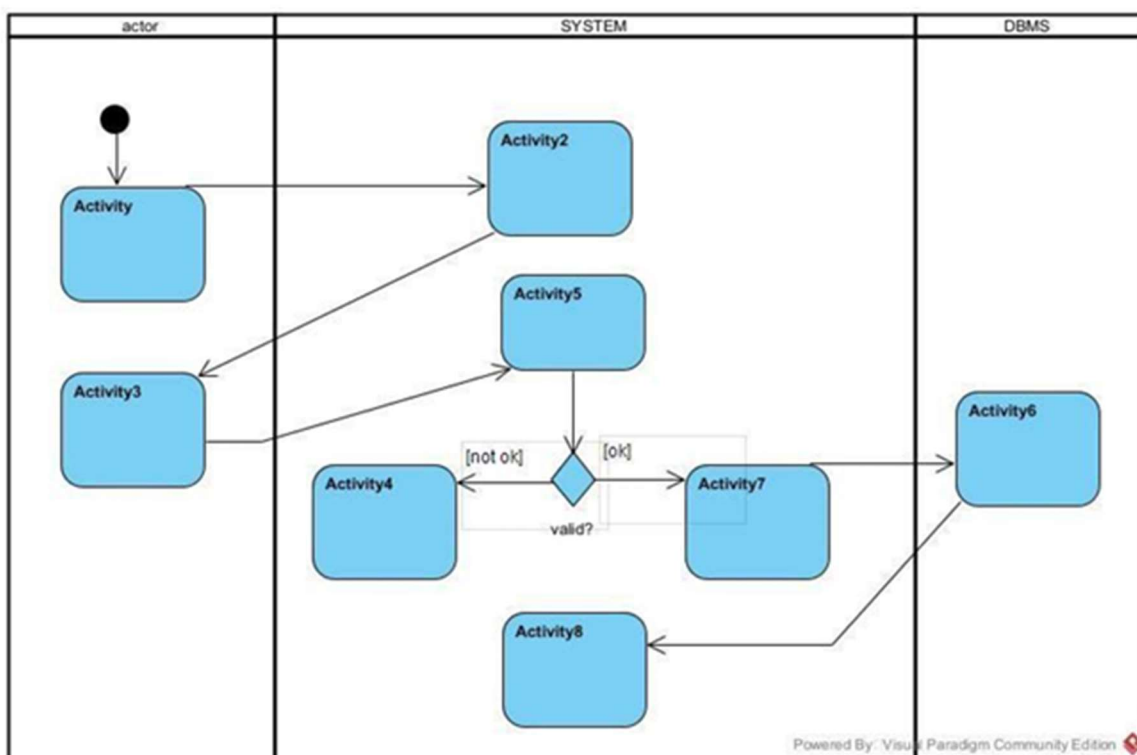
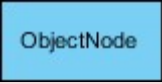
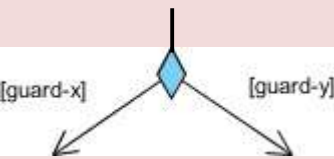
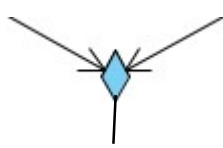
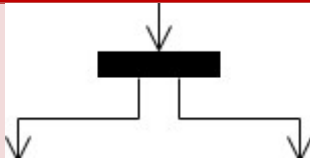
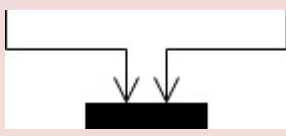
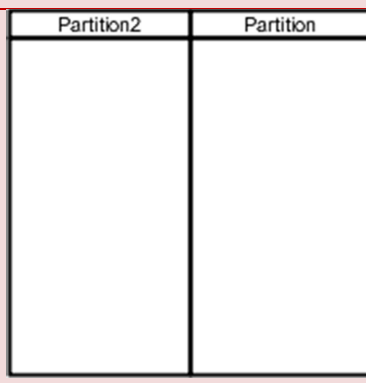


Figure 17 Formalism of activity diagram

iii. COMPONENTS OF ACTIVITY DIAGRAM

Table 19 Components of activity diagram

Element	Notation	Description
Activity		Is used to represent a set of actions
Action		A task to be performed
Activity Edge		An activity edge in a UML activity diagram is a directed connection between two activity nodes. It represents the flow of control or data between these nodes.
Initial node		Portrays the beginning of a set of actions or activities
Activity Final Node		Stop all control flows and object flows in an activity (or action)
Object node		Represent an object that is connected to a set of Object Flows
Decision node		Represent a test condition to ensure that the control flow or object flow only goes down one path

Merge node		Bring back together different decision paths that were created using a decision-node.
Fork node		Split behavior into a set of parallel or concurrent flows of activities (or actions)
Join node		Bring back together a set of parallel or concurrent flows of activities (or actions).
Swimlane and Partition		A way to group activities performed by the same actor on an activity diagram or to group activities in a single thread

iv. ACTIVITY DIAGRAM OF AUTHENTICATE

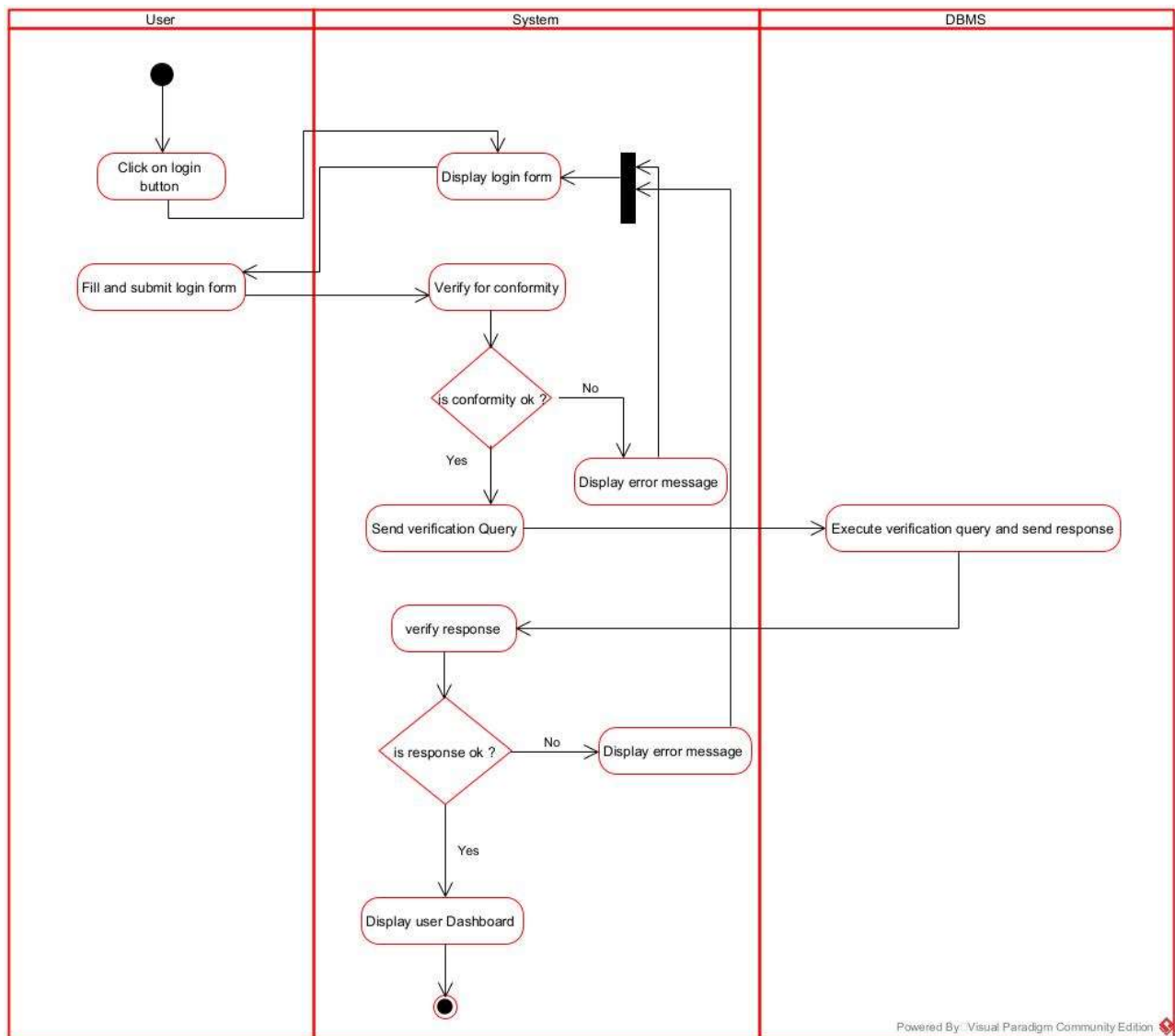


Figure 18 Authenticate activity diagram

v. MAKE GENOTYPE TEST MATCH ACTIVITY DIAGRAM

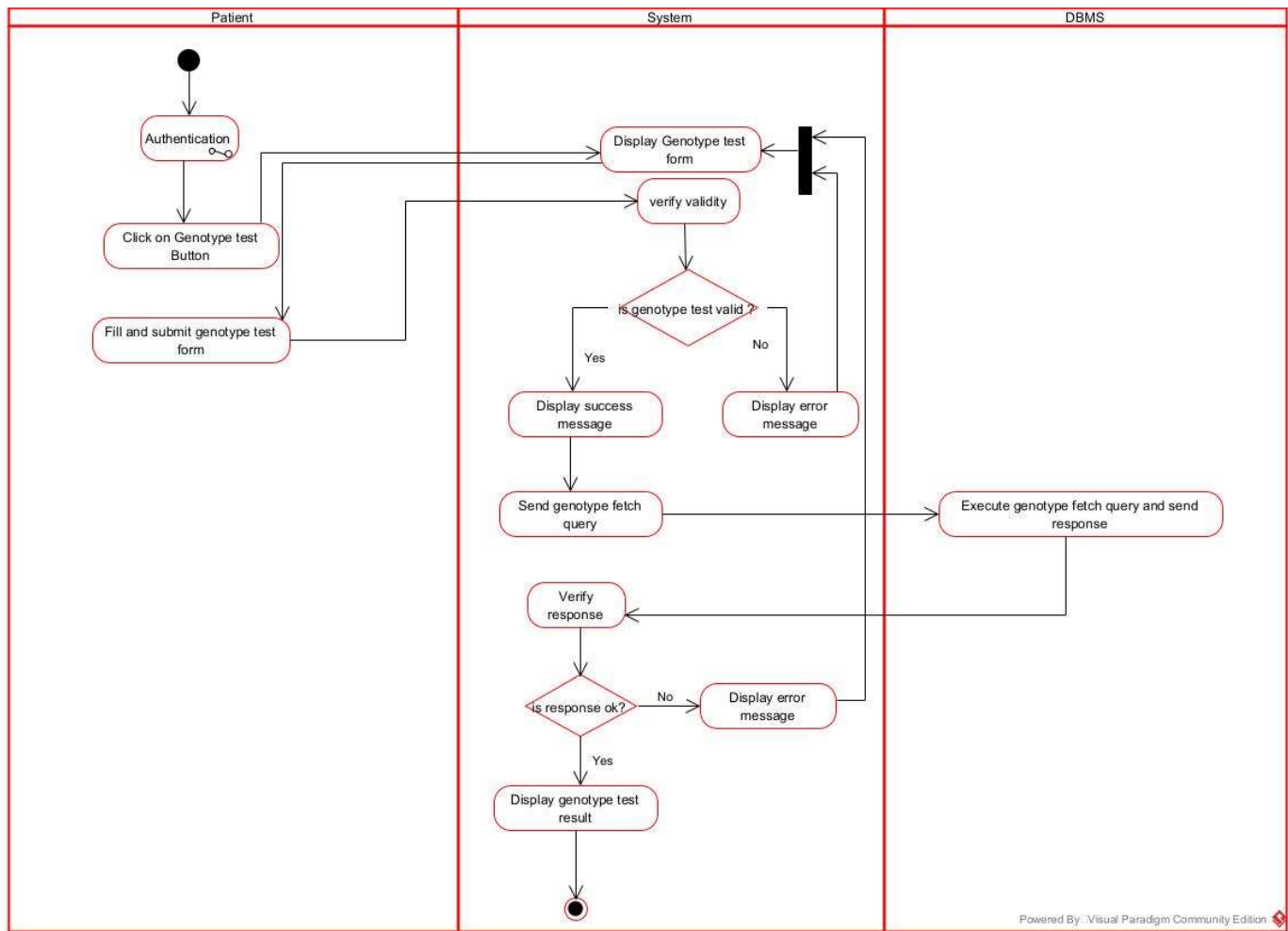


Figure 19 make genotype test match activity diagram

vi. MAKE PAYMENT ACTIVITY DIAGRAM

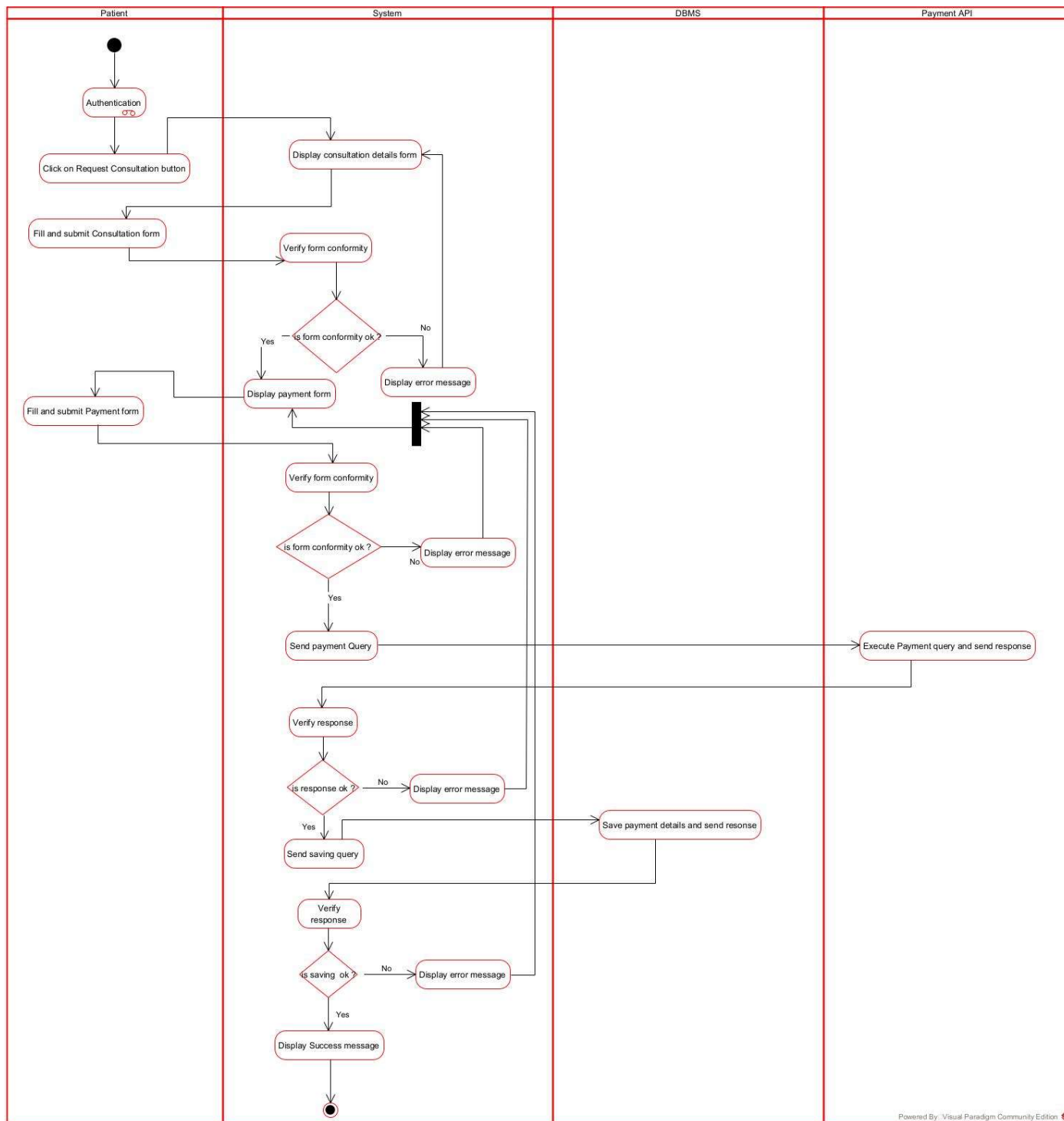


Figure 20 make payment activity diagram

CONCLUSION

Having reached the end of our analysis, we were asked to illustrate the language and method used during this phase, present the different diagrams of the functional branch. we can then say that this analysis file has allowed us delimit the functional needs of the web application that will be designed and to have a detailed overview of the new system to be set up without notwithstanding we quickly dive into the conception phase.



FILE 4 : CONCEPTION PHASE

Preamble

This document makes it possible to model the proposed solution as a whole and to gather the information necessary to set up a complex and efficient database. It therefore foresees the future system.

Content

INTRODUCTION

- I. CAPTURE OF TECHNICAL NEEDS
- II. CLASS DIAGRAM
- III. STATE MACHINE DIAGRAM
- IV. PACKAGE DIAGRAM

CONCLUSION

INTRODUCTION

The realization of an analysis file is very important for the advancement of the application, it guarantees the reliability of maintenance and use of a software. That said, after analyzing our system, we will move on to the design of our future system. To do this we will follow the steps of the technical branch of the 2TUP process, making the diagrams associated with these steps namely the class diagram and the state machine diagram.

I. CAPTURE OF TECHNICAL NEEDS

i. PHYSICAL ARCHITECTURE

We implemented a three-tier architecture system.

The three-tier architecture is a client-server software architecture that consists of the user interface (presentation tier), business rules (logic tier), and data access/storage tier (data tier).

- ❖ **Presentation tier:** It is the user interface and communication layer of the application, where the end user interacts with the application. Its main purpose is to display information to and collect information from the user.
- ❖ **Logic tier:** The logic tier, also known as the application tier or middle tier, is the heart of the application. In this tier, information collected in the presentation tier is processed sometimes against other information in the data tier using business logic, a specific set of business rules. The logic tier can also add, delete or modify data in the data tier.
- ❖ **Data tier:** The data tier, sometimes called database tier, data access tier or backend, is where the information processed by the application is stored and managed. This can be a relational database management system such as PostgreSQL, MySQL, MariaDB, Oracle, DB2, Informix or Microsoft SQL Server, or in a NoSQL Database server such as Cassandra, CouchDB or MongoDB. In a three-tier application, all communication goes through the logic tier. The presentation tier and the data tier cannot communicate directly with one another.

ii. LOGICAL ARCHITECTURE

The logical architecture of an application refers to the structural design that gives as much detail as possible without defining the system technology or environment. The architecture we have chosen for our application is the Model-View- Controller (MVC). The MVC architecture divides an interactive application into 3 components which are:

The model: this component corresponds to all the data-related logic of the application. It directly manages the data, logic and rules of the application.

The view: It is used for all the UI logic of the application. It obtains its data from the model via the controller and each view has an association controller component.

The controller: it acts as an interface between Model and View components to process all the business logic and incoming requests, manipulate data using the Model component and interact with views to render final output. The MVC can be represented as follows:

MVC Architecture Pattern

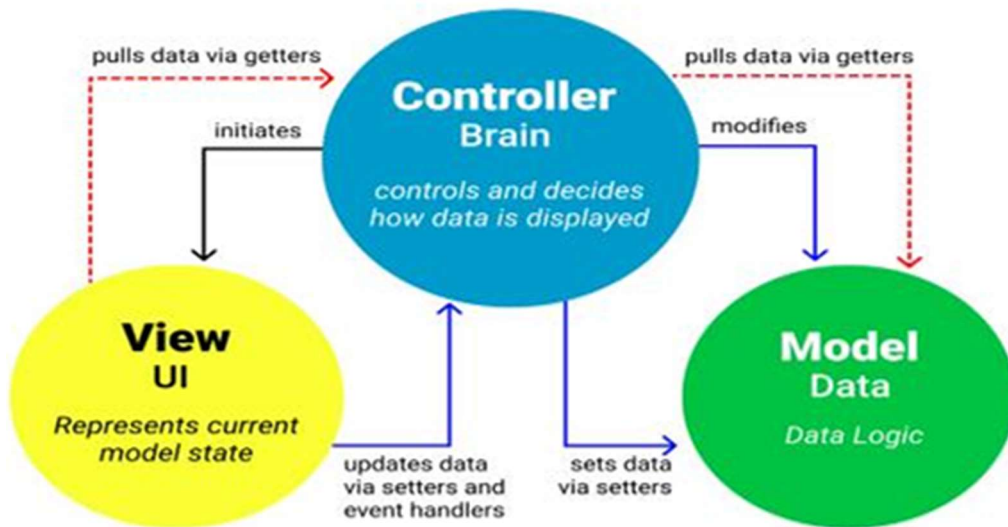


Figure 21 MVC architecture pattern

II. CLASS DIAGRAM

i. DEFINITION

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing, and documenting different aspects of the system, but also for constructing executable code of the software application. A class diagram describes the attributes and operations of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

ii. FORMALISM

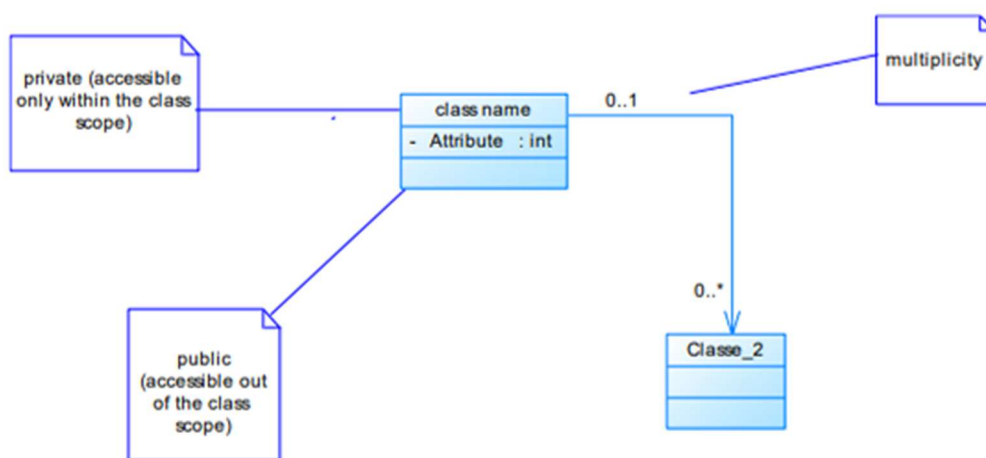
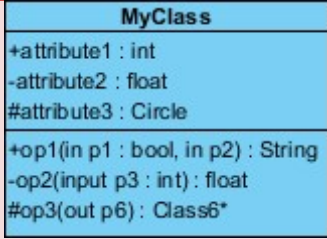
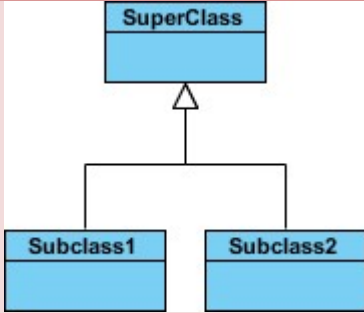
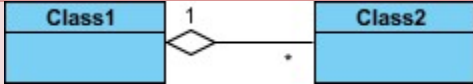


Figure 22 Formalism of a class diagram

iii. COMPONENTS OF CLASS DIAGRAM

Table 20 Components of a class diagram

Element	Notation	Description
Class		A class is an element that defines the attributes and behaviours that an object can generate
Generalization		it a relationship between a whole thing (called superclass) and a more specific thing (called subclass)
Association		It is a general type of relationship between elements, it may include cardinality, roles etc.
Aggregation		If the parent of the aggregate is deleted, the children are not deleted.
Composition		If a parent of a composite is deleted all its parts are deleted with it.
Dependency		It existed between two classes, if one changes it may cause the change in the order, but the other way around

iv. CLASS DIAGRAM OF DREPALIFE

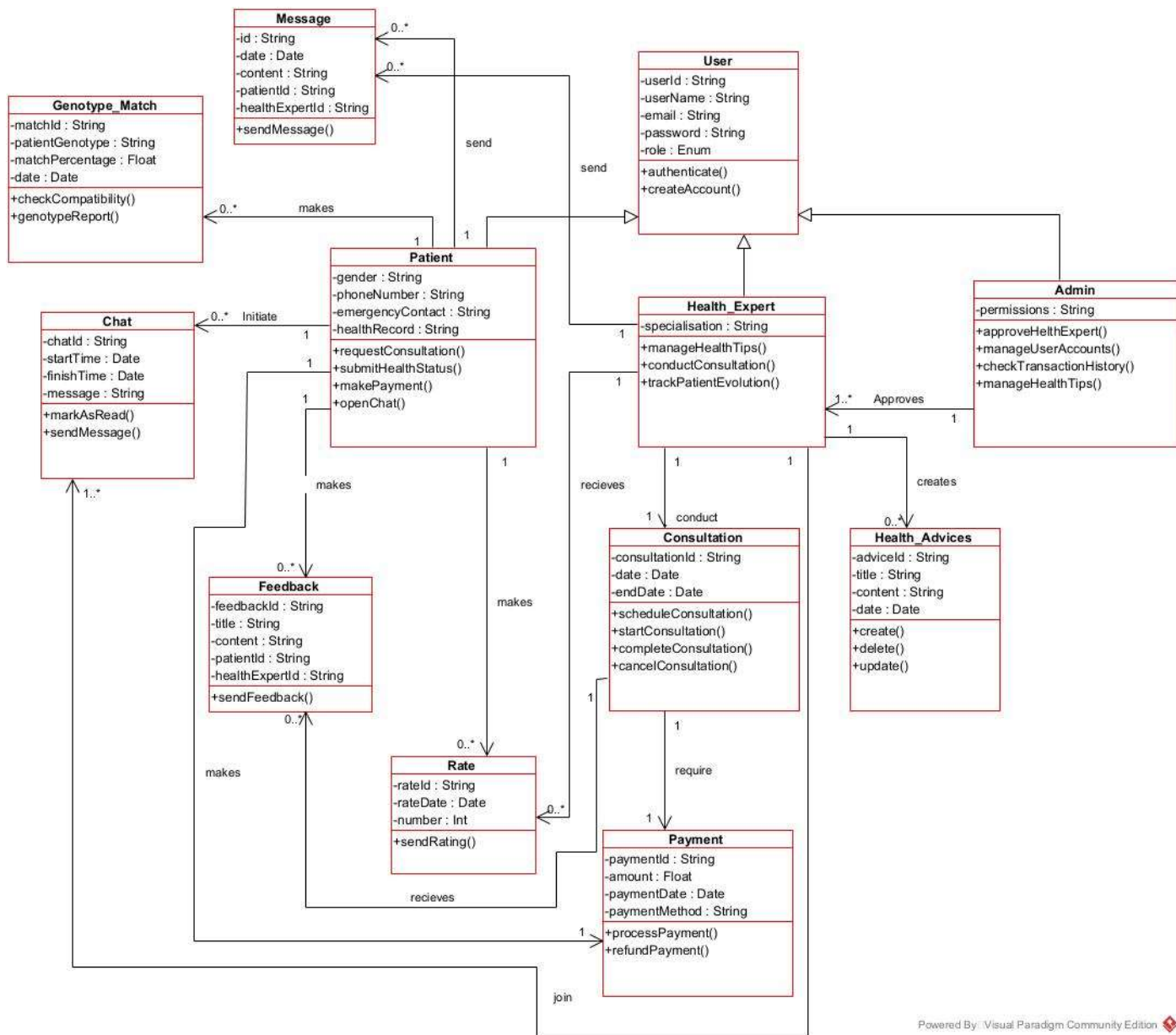


Figure 23 Drepalife Class Diagram

III. STATE MACHINE DIAGRAM

i. DEFINITION

A state machine diagram describes the behavior of a single object in response to a series of events in a system. Also known as the state machine diagram, it models the dynamic flow of control from the state of a particular object within a system.

ii. FORMALISM

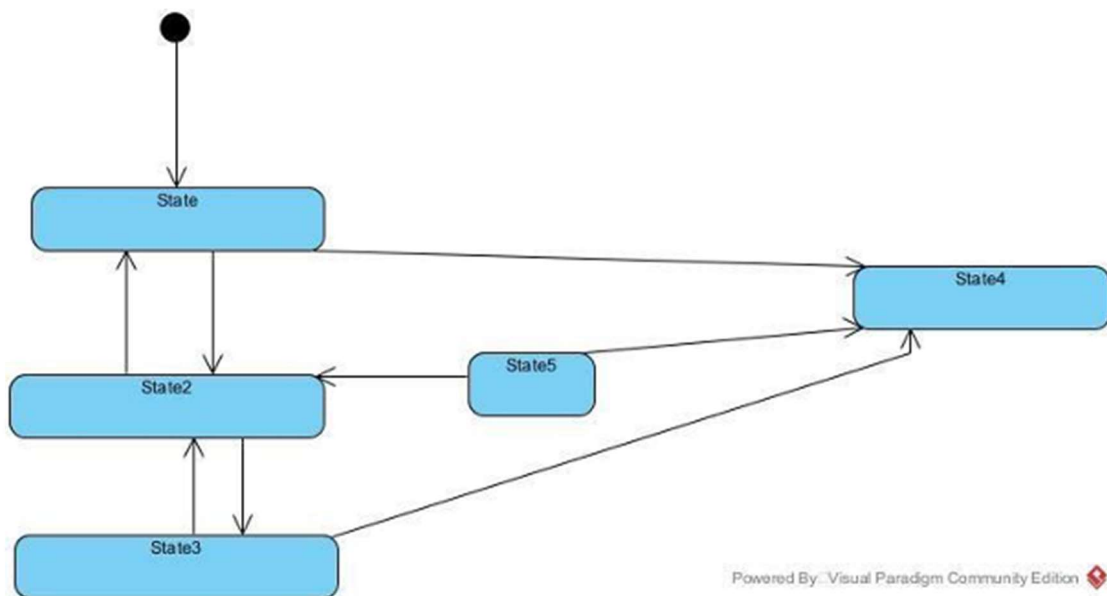
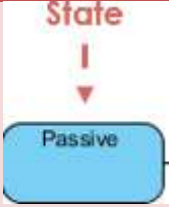
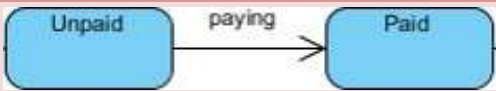


Figure 24 Formalism of state machine diagram

iii. COMPONENTS OF STATE MACHINE DIAGRAM

Table 21 Components of a state machine diagram

Element	Notation	Description
State		Models a situation during which a certain invariant condition holds.
First (Initial State)		It represents a default vertex, that is, a source for a single transaction to the default or composite state.
Final state		A state specifying that the enclosing region is complete
Transition		A direction relation between a source and a target vertex.
Choice pseudo state		A diamond symbol that indicates a dynamic condition with branched potential results
Terminate		Implies that the execution of a state by means of its context is terminated.

iv. ACCOUNT STATE MACHINE DIAGRAM

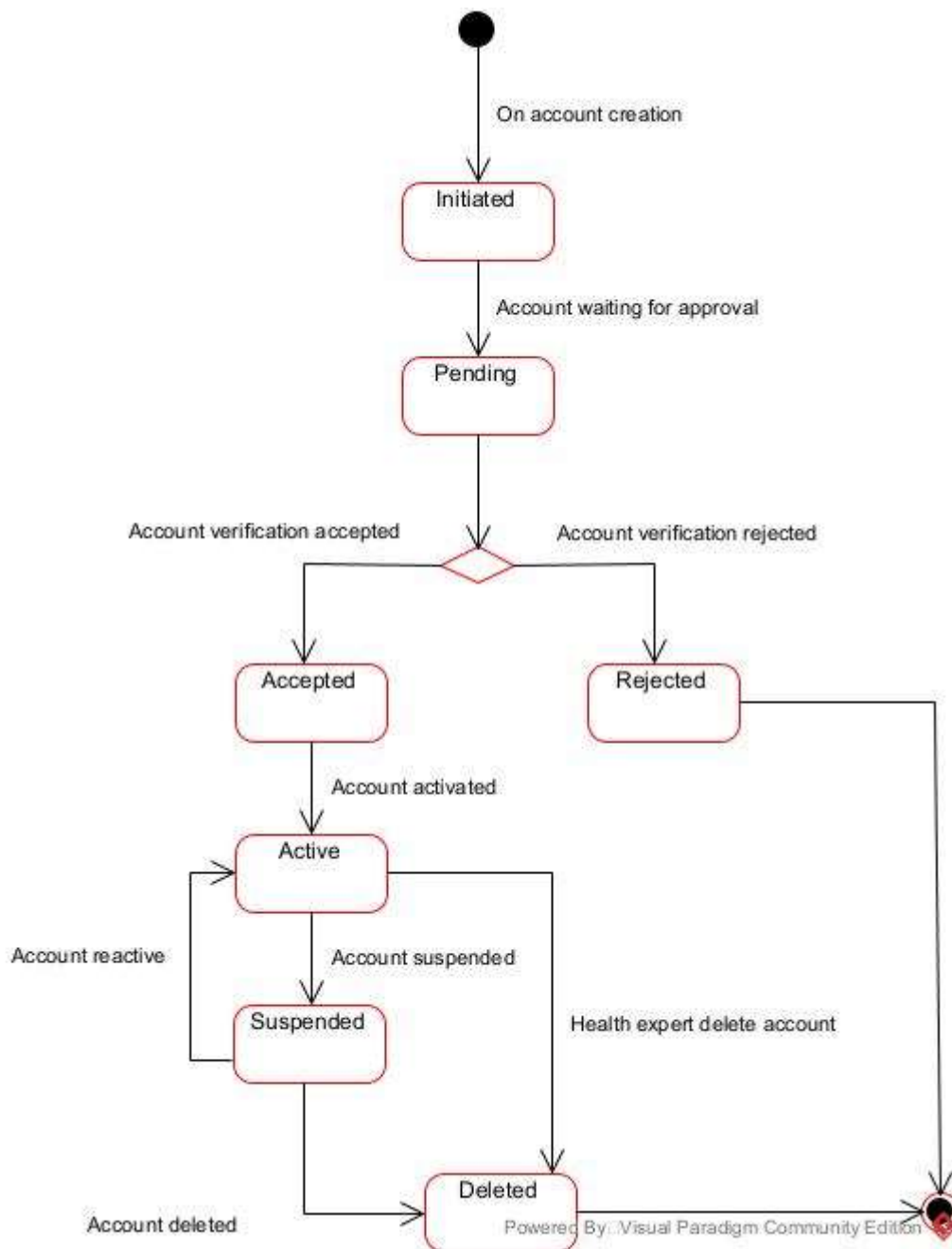


Figure 25 Account state machine diagram

IV. PACKAGE DIAGRAM

i. DEFINITION

A package diagram is a type of structural diagram in UML that represents the organization and dependencies of different packages within a system or software application. Package diagrams are useful for visualizing the overall structure and organization of a system, as well as understanding the dependencies and relationships between the different components.

ii. FORMALISM

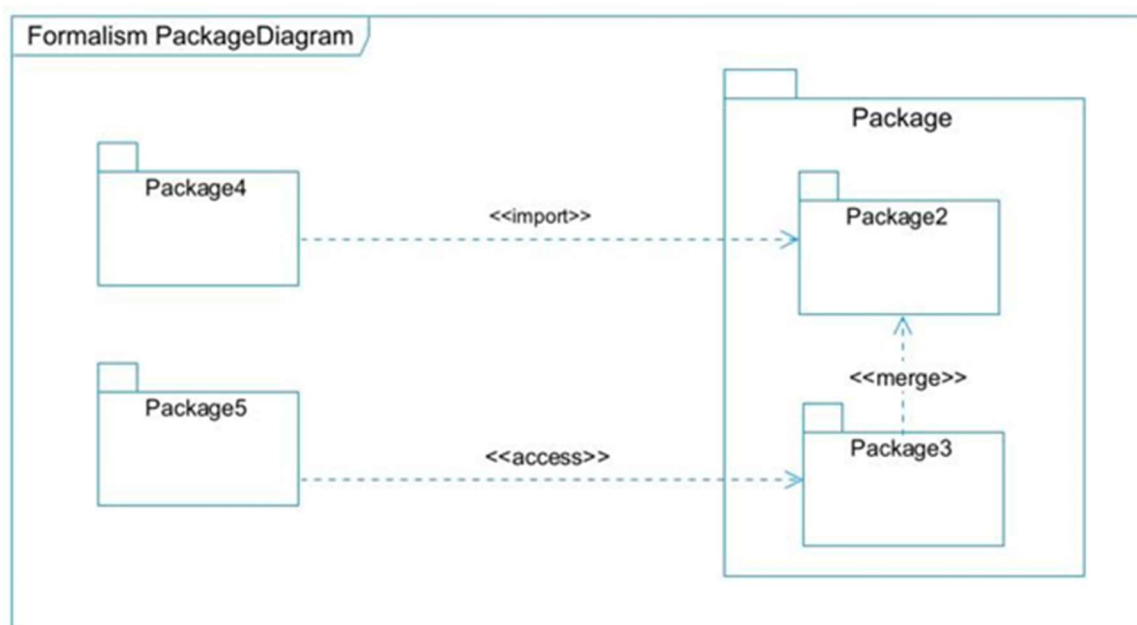
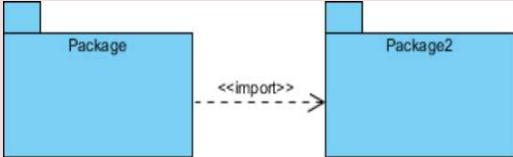
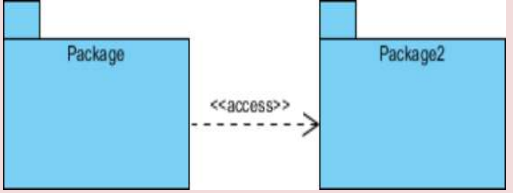


Figure 26 Formalism of package diagram

iii. COMPONENTS OF PACKAGE DIAGRAM

Table 22 Components of a package diagram

Element	Notation	Description
Package		A package is a namespace use to group related elements; it is a mechanism used to group elements into a better structure in a system.
Package import		A relationship Indicate that, functionality has been imported from one package to another.
Package access		A relationship Indicates that one package requires assistance from the function of another package.
Package merge		It is a relationship which shows that, the functionality of two packages are combines to a single function.

iv. PACKAGE DIAGRAM OF DREPALIFE

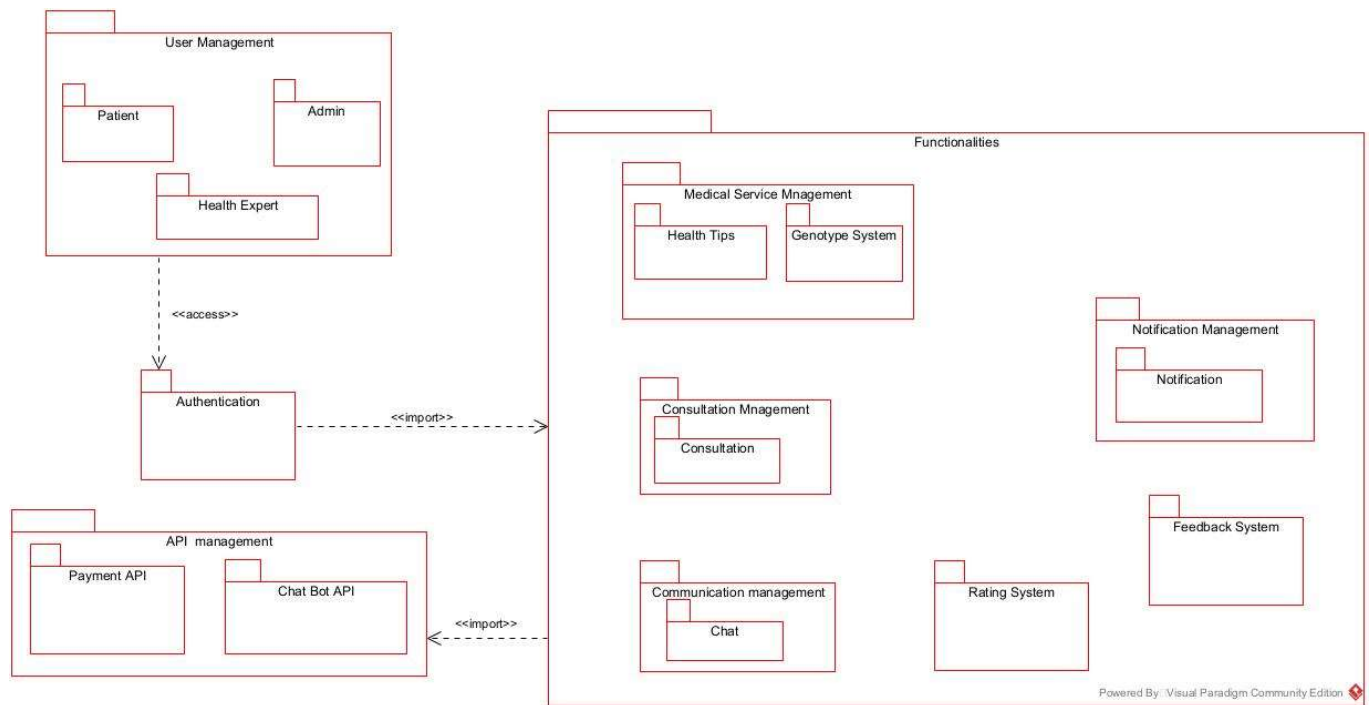


Figure 27 Drepalife package diagram



CONCLUSION

In the conception phase, we set as objective to plans the different aspects of our system by showing how it will be structured and deployed within existing technical architectures. We began by considering the technical constraints for our system, after which we proceeded to identify the components of our system, how they are grouped together and how they should be deployed on appropriate deployment targets. We finished this phase by looking at interactions between the various aspects and actors of our system. The next phase of our report is the realization phase where we will look at aspects concerning the implementation of our system.

FILE 5 : REALIZATION

Preamble

The realization document consists of the implementation of the project in a programming language according to the specifications defined in the previous file. It contains component and deployment diagrams, the testing and development phases. At the end of this part, programming documentation will be produced explaining the architecture of the database and the architecture of our code.

Content

INTRODUCTION

- I. DEPLOYEMENT DIAGRAM
- II. COMPONENT DIAGRAM
- III. TECHNICAL CHOICES

CONCLUSION

INTRODUCTION

Design and modeling are important links in the development process of an application because they make it possible to define and specify the different elements constituting the application or system studied. Realization is the culmination of the work carried out upstream because it makes it possible to physically create the various objects that were designed, and to bring the computer system to life. Throughout this phase, we will present the software, physical, and hardware architecture of our system through diagrams such as the component diagram, deployment diagram, physical model, and code architecture.

I. DEPLOYMENT DIAGRAM

i. DEFINITION

The deployment diagram models the hardware components used to implement a system and the association between those components. Deployment diagrams can be implemented as early as the design phase to document the system's physical architecture. As elements used in deployment diagrams, we can have, components, as in component diagrams, and nodes, which represent the physical processing resources of the system, and their associations.

ii. FORMALISM

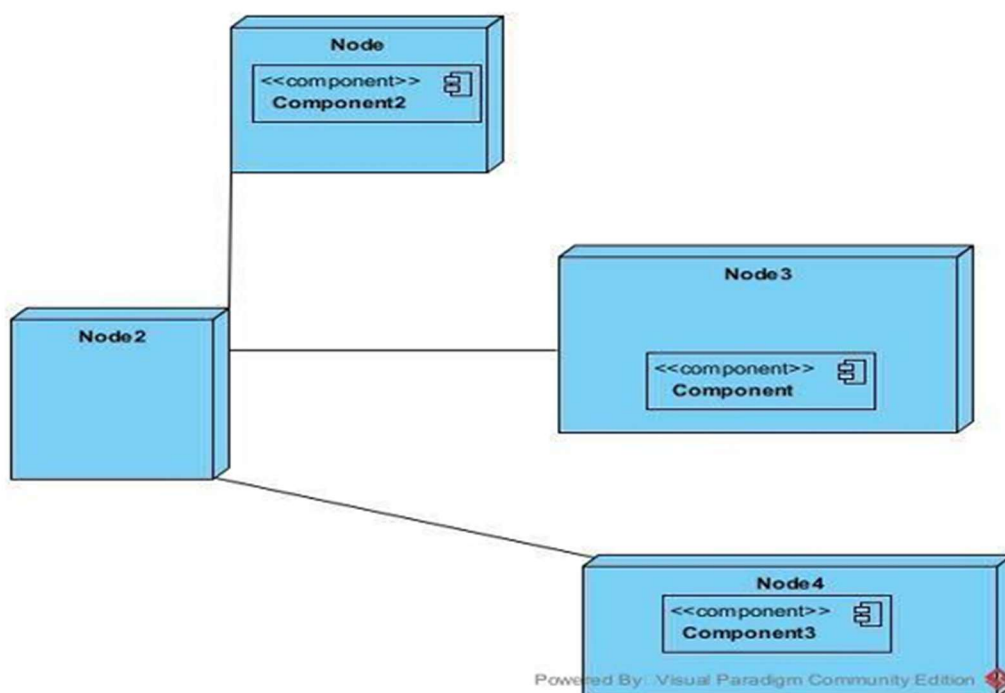


Figure 28 Formalism Deployment diagram

iii. COMPONENTS OF DEPLOYMENT DIAGRAM

Table 23 Component of the deployment diagram

Element	Notation	Description
Node		A hardware or software object, shown by a three-dimensional box
Artifact		An artifact is a major product, which is produced or used during the development of a software. E.g diagrams, data models, setup scripts
Component	 component 1	It represents a modular part of a system that encapsulates its content and whose manifestation is replaceable within its environment.
Association	association	An association helps to connect two nodes together which permits them to communicate together

iv. DEPLOYMENT DIAGRAM OF DREPALIFE

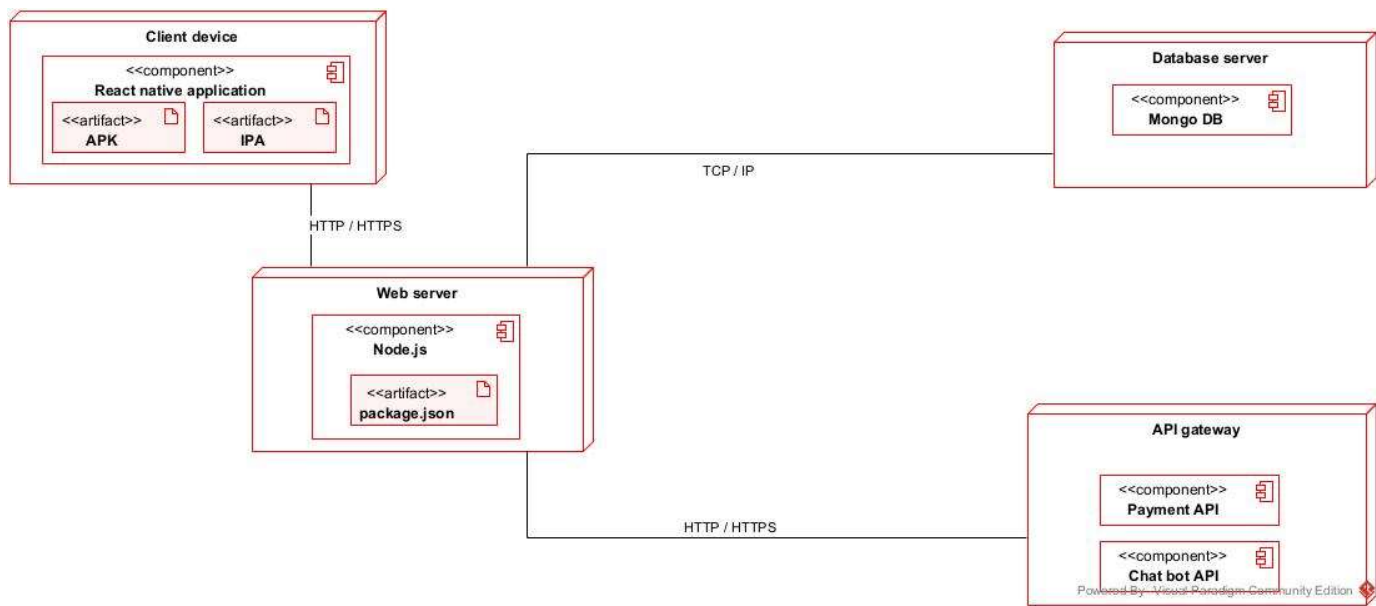


Figure 29 Drepalife Deployment diagram

II. COMPONENT DIAGRAM

i. DEFINITION

Component diagrams are used to model the physical aspect of a system. Now the question is, what is this physical aspect? They are elements such as Executables, libraries, files, documents, etc., which reside in a node. The component diagram does not describe the *functionality of the system*, but it describes the components used to make those functionalities.

ii. FORMALISM

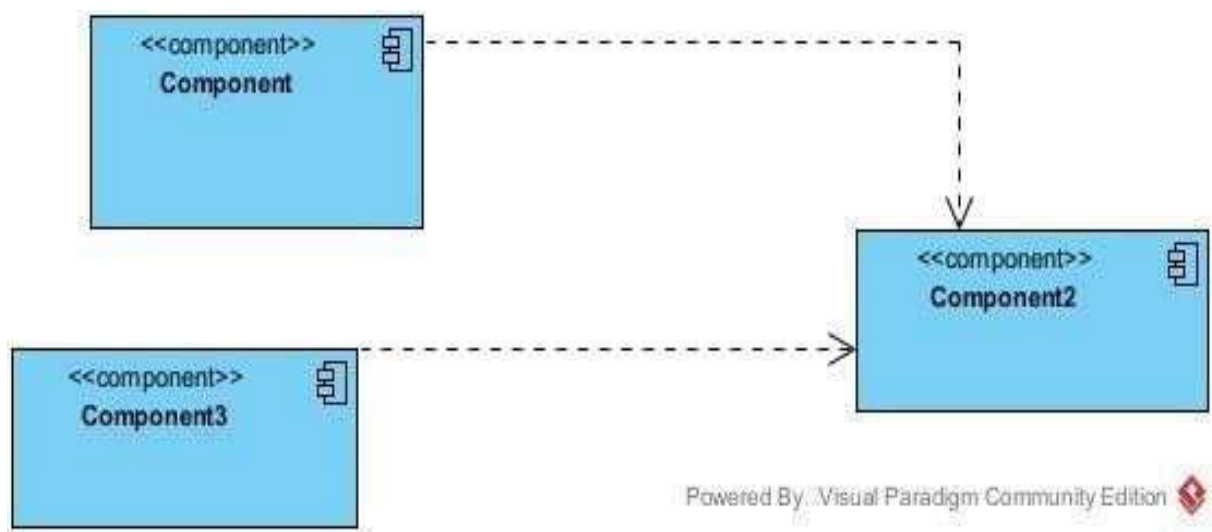
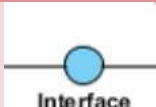
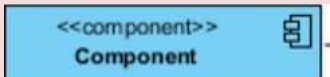


Figure 30 Formalism of Component diagram

iii. COMPONENTS OF COMPONENT DIAGRAM

Table 24 Components of component diagram

Element	Notation	Description
Dependency		A dependency is a relationship that signifies that a single or a set of model elements requires other model elements for their specification or implementation.
Port		A port is often used to help expose required and provided interfaces of a component.
Interface		It is a point of interaction between a component and its environment or other components. It specifies the services that a component provides or requires from other components
Component		It represents a self-contained and replaceable unit that interacts with other components via provided and required interfaces

iv. COMPONENT DIAGRAM OF DREPALIFE

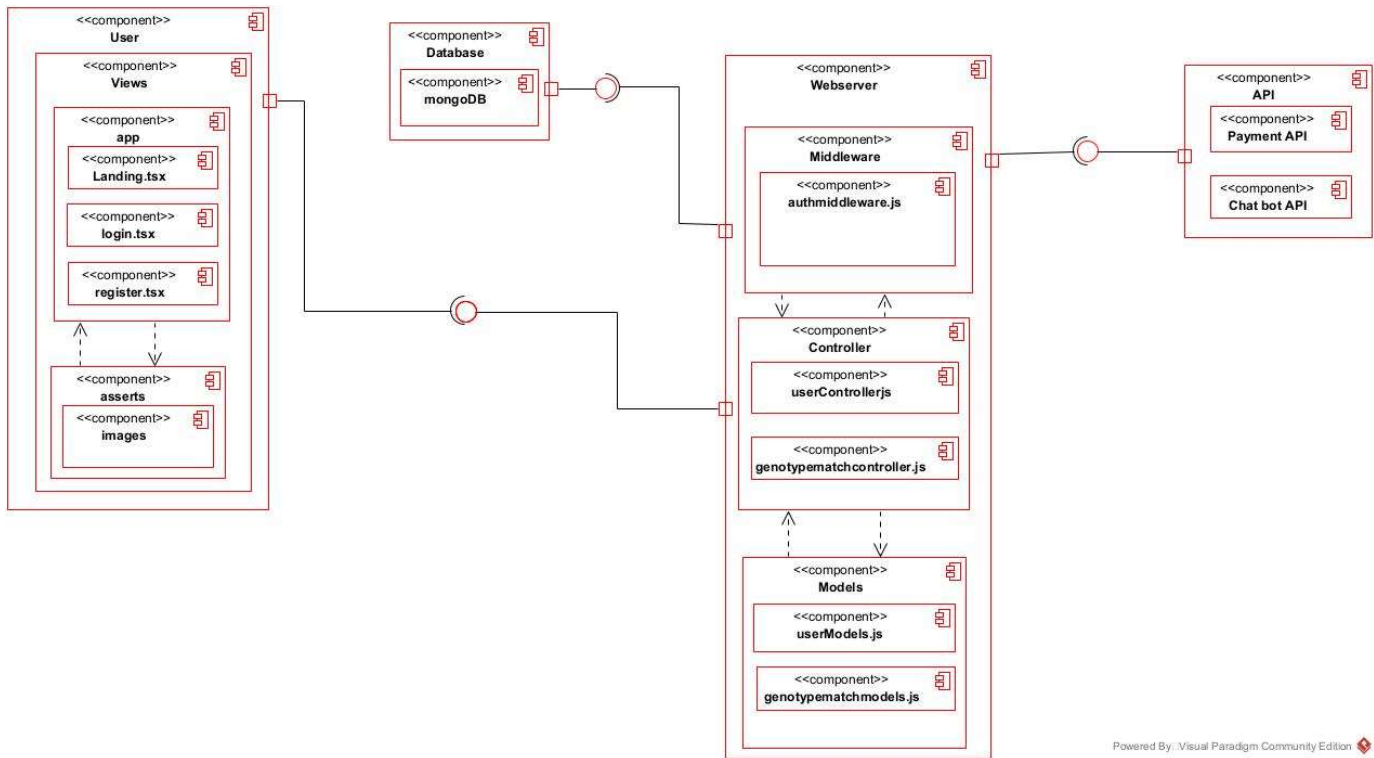


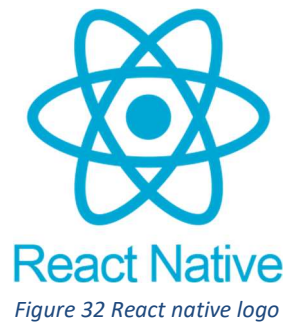
Figure 31 Drepalife Component diagram

III. TECHNICAL CHOICES

To set up our platform and meet the needs of the end-users, the technologies chosen are React Native for the front-end, combined with Node.js for the back-end, and MongoDB for interaction with the database. Each of these technologies will be presented in detail in the following paragraphs.

a. React Native;

React Native is an open-source framework. It allows developers to build mobile applications using JavaScript. In software development, a framework serves as a platform for developing applications. It contains resources like pre-built components, libraries, and reference material. It enables the creation of cross-platform apps that can run on both iOS and Android using a single codebase, which saves development time and effort.



b. Node js;

Node.js is an open-source, cross-platform JavaScript runtime environment that allows developers to run JavaScript code on the server side. It uses Chrome's V8 JavaScript engine to execute code and is designed to build scalable and high-performance applications, especially for web services and APIs.



Figure 33 node js logo

c. MongoDB;

MongoDB is an open source, cross-platform, document-oriented database program. Classified as a NoSQL database product, MongoDB utilizes JSON-like documents with optional schemas. MongoDB is developed by MongoDB Inc.



Figure 34 mongoDB logo

d. Gantt Project;

Project Gantt is a widely used project management tool for planning and tracking project tasks over a given period. A Gantt chart is particularly useful for visualizing dependencies between tasks, start and finish dates, and project progress.

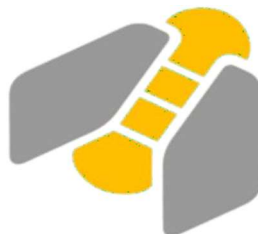


Figure 35 Gantt Project

e. Visual paradigm;

Visual Paradigm Community Edition is a UML software that supports all UML diagram types. It is an international award-winning UML modeler, and yet it is easy-to-use, intuitive & completely free.



Figure 36 visual paradigm logo

f. Postman

Postman is a popular API development tool that facilitates the creation, testing and documentation of APIs. It is widely used by developers, testers and API development teams to simplify the process of working with HTTP APIs.



Figure 37 postman logo

CONCLUSION

During our realization phase, we successfully developed and implemented our application. To achieve this, we utilized our analysis and conception phase. We also created deployment and component diagrams which illustrate the modules, files, and assets, as well as how different elements interact with each other within our system. Our next step is to move onto the functionalities testing phase, where we will assess the efficacy of the various modules present in our app for different users.



FILE 6 : FUNCTIONALITY TESTING

Preamble

This phase aims at testing the software or application to verify if the system is working as expected. It involves testing individual functionalities and capabilities of the software to ensure that they work correctly and meet the specified needs.

Content

INTRODUCTION

I. APPLICATION FUNCTIONALITIES

II. TEST

CONCLUSION

INTRODUCTION

It is a systematic process that involves evaluating the functionality of a software application or system in different phases or stages of its development lifecycle. This testing approach ensures that each phase of the software development process is thoroughly tested for its intended functionalities and overall performance. Its main goal is to identify and rectify any defects and bugs during the process.



I. APPLICATION FUNCTIONALITIES

➤ **Authentication**

This functionality enables the users to have access to their workspace or dashboard. In case he/she does not have an account he/she will register.

➤ **Genotype Compatibility Check**

Users input their genotype with that of a partner to receive instant risk predictions for their future children.

➤ **Consultation Booking & Payments** Families can securely book and pay for consultations with verified providers through the app.

➤ **Chatbot Assistance**

A conversational assistant answers common questions, guides users through risk checks, and directs them to educational or medical support.

➤ **Join Community**

This functionality enables the patient to join communities for better knowledge of the sickness

➤ **Account and Management**

Admin is capable of creating, deleting, blocking, or suspending accounts

II. TEST

a. DEFINITION

Testing, in simple terms, involves verifying that your code functions correctly. It is important because it boosts your confidence in the code and ensures its reliability. There are different types of testing, including:

- **Unit test:** In a unit test, you are testing whether each function is working properly.
- **Feature test:** In a feature test, you are testing whether your routes, controllers, and database behave as expected.
- **Manual testing:** Here, you are manually testing your application.
- **Alternative or Automated testing:** Using online tools to test your code.

b. TESTS MADE IN THE SCOPE OF OUR PROJECT

For testing the functionality of DrepaLife, we used Jest, a powerful JavaScript testing framework that allows us to perform unit and integration tests on both frontend and backend components. Jest ensures that each module of the application behaves as expected, helping to catch errors early and maintain the reliability of the platform.

```
Tests: 1 failed, 1 total
Snapshots: 0 total
Time: 4.662 s, estimated 6 s
Ran all test suites.

dell@DESKTOP-KC0Q5RQ MINGW64 ~/Desktop/Drepalife/drepalife_backend (master)
$ npm test

> backend@1.0.0 test
> jest --runInBand --config jest.config.cjs

(node:19028) [MONGODB DRIVER] Warning: useNewUrlParser is a deprecated option: useNewUrlParser has no effect since Node.js
Driver version 4.0.0 and will be removed in the next major version
(Use `node --trace-warnings ...` to show where the warning was created)
(node:19028) [MONGODB DRIVER] Warning: useUnifiedTopology is a deprecated option: useUnifiedTopology has no effect since No
de.js Driver version 4.0.0 and will be removed in the next major version
PASS test/integration/auth.test.js
  Authentication Testing (Register and Login)
    ✓ Signup: successfully creates a user and returns 201 without password (426 ms)

Test Suites: 1 passed, 1 total
Tests: 1 passed, 1 total
Snapshots: 0 total
Time: 4.751 s, estimated 5 s
Ran all test suites.
```

Figure 38 jest authentication test result

```
import request from 'supertest';
import mongoose from 'mongoose';
import { MongoMemoryServer } from 'mongodb-memory-server';
import app from '../app.js';

describe('Authentication Testing (Register and Login)', () => {
  let mongoServer;

  beforeAll(async () => {
    mongoServer = await MongoMemoryServer.create();
    const mongoUri = mongoServer.getUri();

    await mongoose.connect(mongoUri, {
      useNewUrlParser: true,
      useUnifiedTopology: true,
    });
  });

  afterAll(async () => {
    await mongoose.disconnect();
    await mongoServer.stop();
  });

  beforeEach(async () => {
    // Clear all collections before each test
    const collections = await mongoose.connection.db.collections();
    for (let collection of collections) {
      await collection.deleteMany({});
    }
  });
});
```

Figure 39 test code interface



CONCLUSION

Testing is an integral part of the software development life cycle and is performed to verify the correctness, reliability, and robustness of the implemented functionality. The functionality and testing phase is therefore a critical stage in software development where the implemented features are validated, and any issues are identified and resolved.



FILE 7 : USER GUIDE

Preamble

A user guide explains how to use a software application in a language that a non-technical person can understand. Thus, it enables the user to easily use the application to familiarize themselves with the software and discover all its functionalities.

Content

INTRODUCTION

I. DESCRIPTION OF THE APPLICATION

II. USER MANUAL

CONCLUSION



INTRODUCTION

The user manual is the document established after the realization of an application, software, or platform. It informs on the question "how to use the software, the application, or the platform that you have in front of you? ". It is therefore crucial for us to offer the various users of our platform a guide allowing them to easily perform various operations on our platform.



I. DESCRIPTION OF THE APPLICATION

Drepalife is a mobile health application designed to raise awareness and reduce early deaths caused by sickle cell disease in Cameroon. It helps families and individuals understand their genotype compatibility, assess sickle cell risks, and access reliable educational resources. The application also connects users to verified healthcare providers and offers a chatbot assistant for quick guidance.

Key features include:

- **User Registration and Authentication:** Create an account, log in, or use the app without authentication.
- **Role Selection:** Identify as a patient or healthcare provider to access tailored features.
- **Genotype Compatibility Test:** Check and understand genotype risks for children.
- **Chatbot Assistant:** Ask questions, get instant answers, and receive step-by-step guidance.
- **Educational Materials:** Access simplified guides and tips on sickle cell prevention and management.
- **Healthcare Provider Access:** View verified testing centers, clinics, and doctors for consultation.
- **Event and Awareness Updates:** Stay informed on campaigns, events, or initiatives related to sickle cell awareness.



II. USER MANUAL

➤ PREREQUISITES

Before using Drepalife, ensure you have:

- An Android smartphone.
- The Drepalife APK file downloaded.
- A stable internet connection (for real-time features such as risk tests and chatbot).

➤ USEFUL STEPS

Below are the steps you will follow to install and use the application:

➤ **Download and Install**

Download the Drepalife APK file.

Open it on your phone and follow the prompts to install.

➤ **Launch the App**

Tap the Drepalife icon to open the application.

➤ **Landing Screen**

Choose to create an account, log in, or continue without authentication.

➤ **Sign Up**

Enter your details (name, email, password, phone, emergency contact, role, gender).

➤ **Log In**

Use your email and password to access your account.

➤ **Role Page**

Select your role (e.g., Patient or Healthcare Provider) to access personalized features.

➤ **Home Page**

View events, announcements, and awareness updates.

➤ **Genotype Test Page**

Input genotypes to get instant compatibility and risk results.

➤ **Chatbot Page**

Interact with the chatbot to ask questions, receive guidance, or navigate the app.

2. SHOW CASE



Figure 41 landing screen

- **Landing Screen:** Create an account, log in, or continue as a guest.

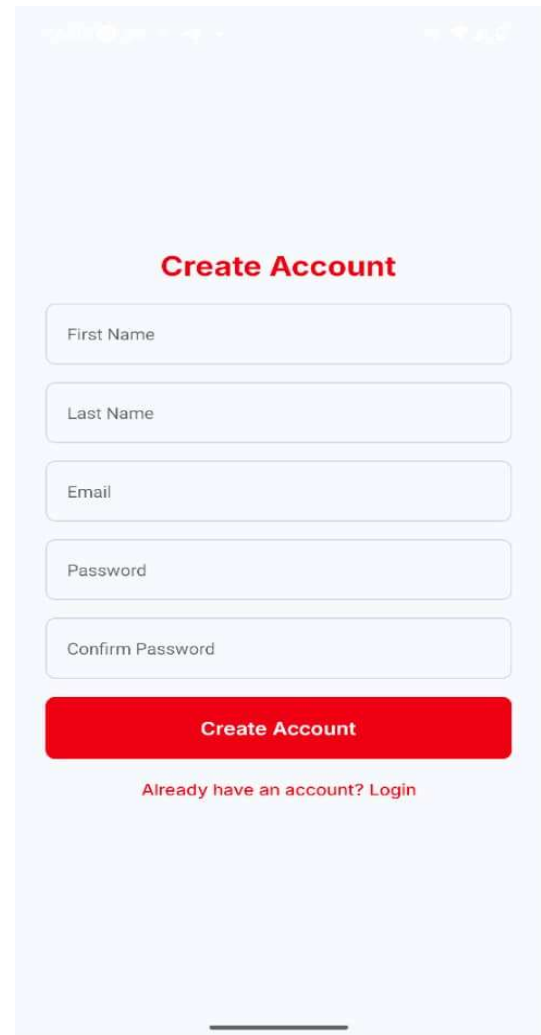
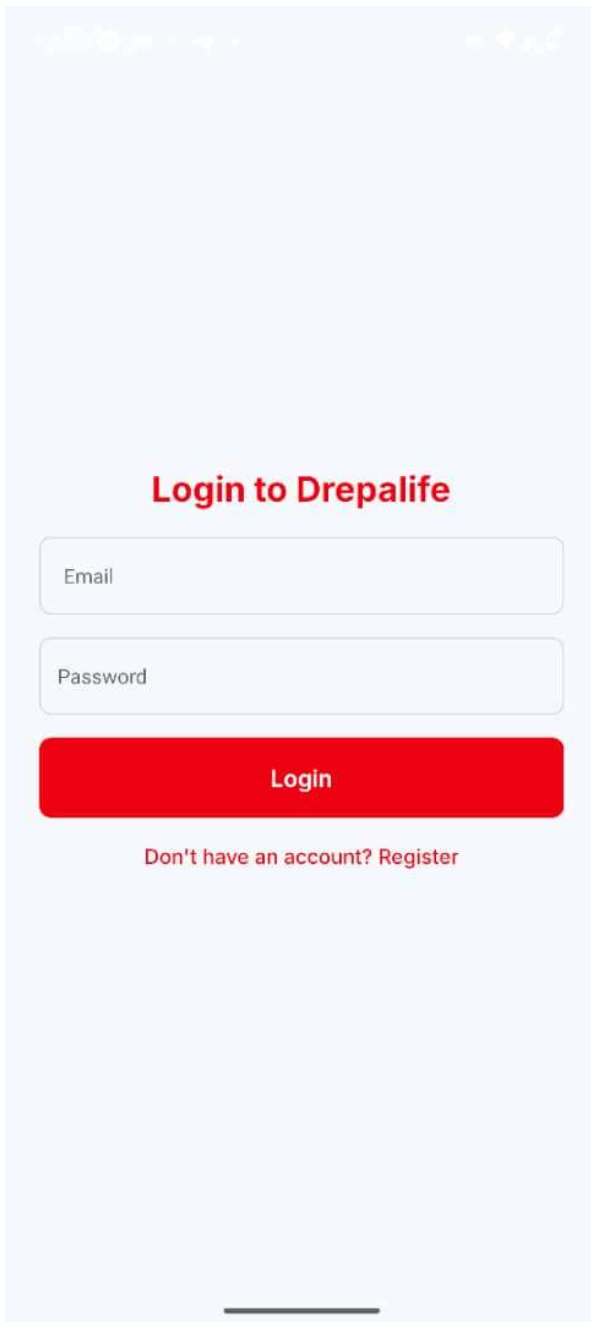


Figure 40 signup screen

- **Sign Up Screen:** Register with personal and health details.



The login screen features a light blue background with a white header area containing social media icons. The main heading is "Login to Drepalife" in red. Below it are two input fields: "Email" and "Password". A red "Login" button is positioned below the password field. At the bottom, there is a link that says "Don't have an account? Register".

Figure 42 login screen



The role selection screen has a white background with a grey header area containing social media icons. The heading is "Select Your Role" in black. There are two red buttons: "Patient" and "Health Expert".

Figure 43 Role screen

- **Log In :** Use your email and password to access your account

- **Role Page:** Select user type for tailored experience.

➤ Patient dashboard

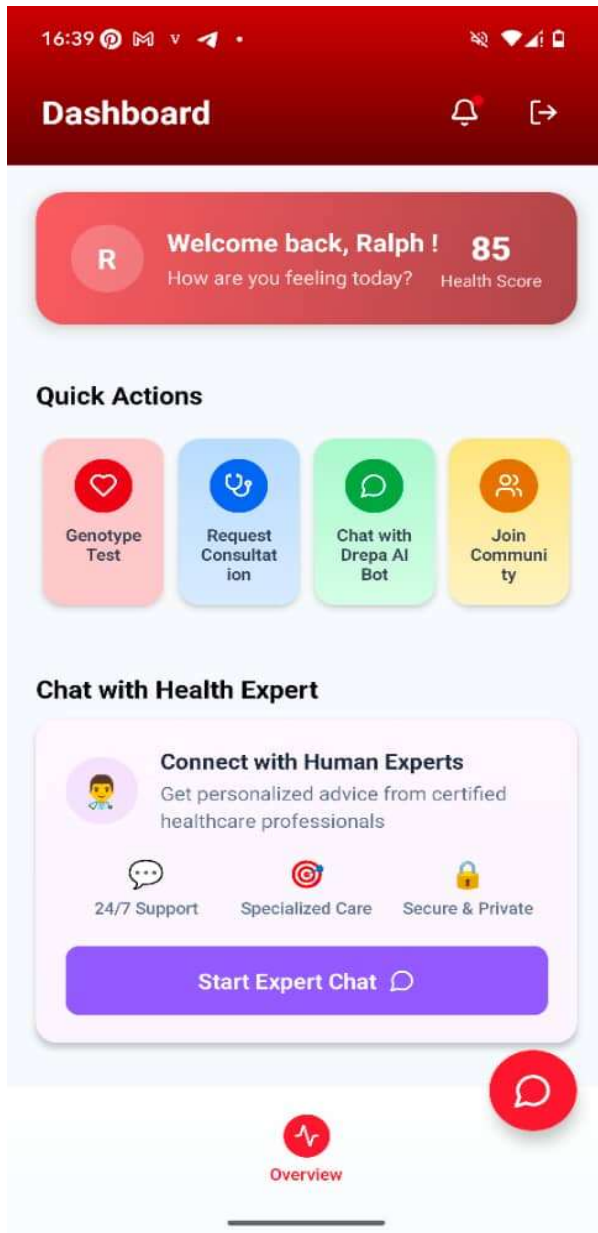


Figure 45 patient dashboard

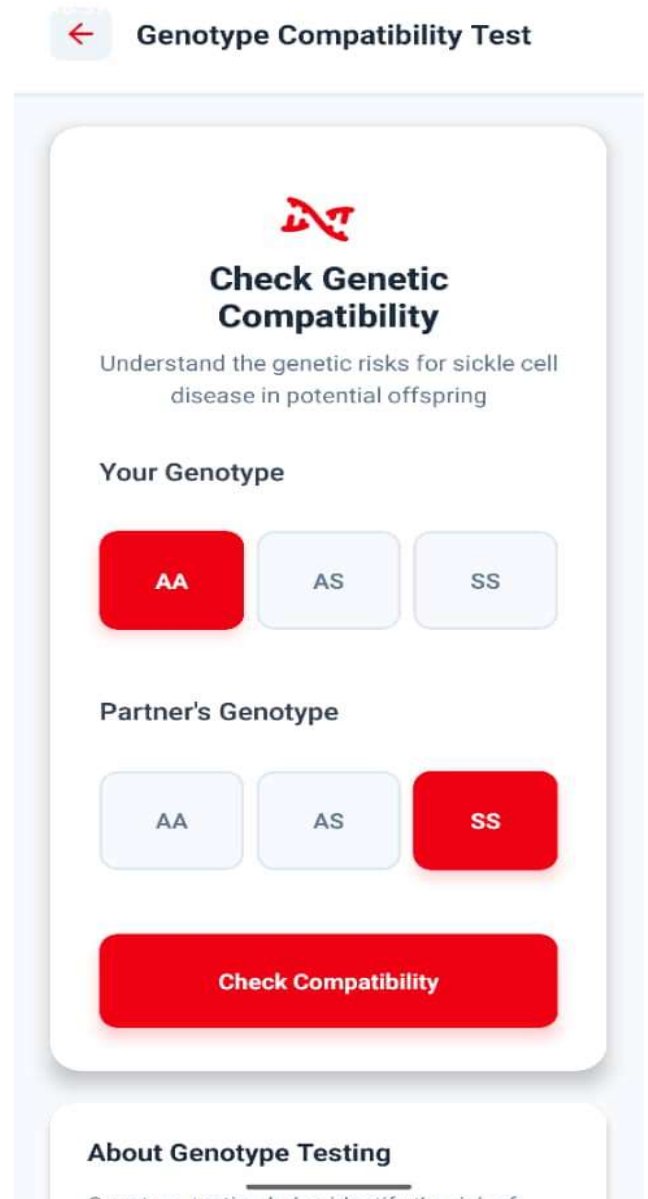


Figure 44 genotype test match screen

- **Genotype Test Page:** Perform genotype compatibility check.

- **Chatbot Page** Access instant support and information.

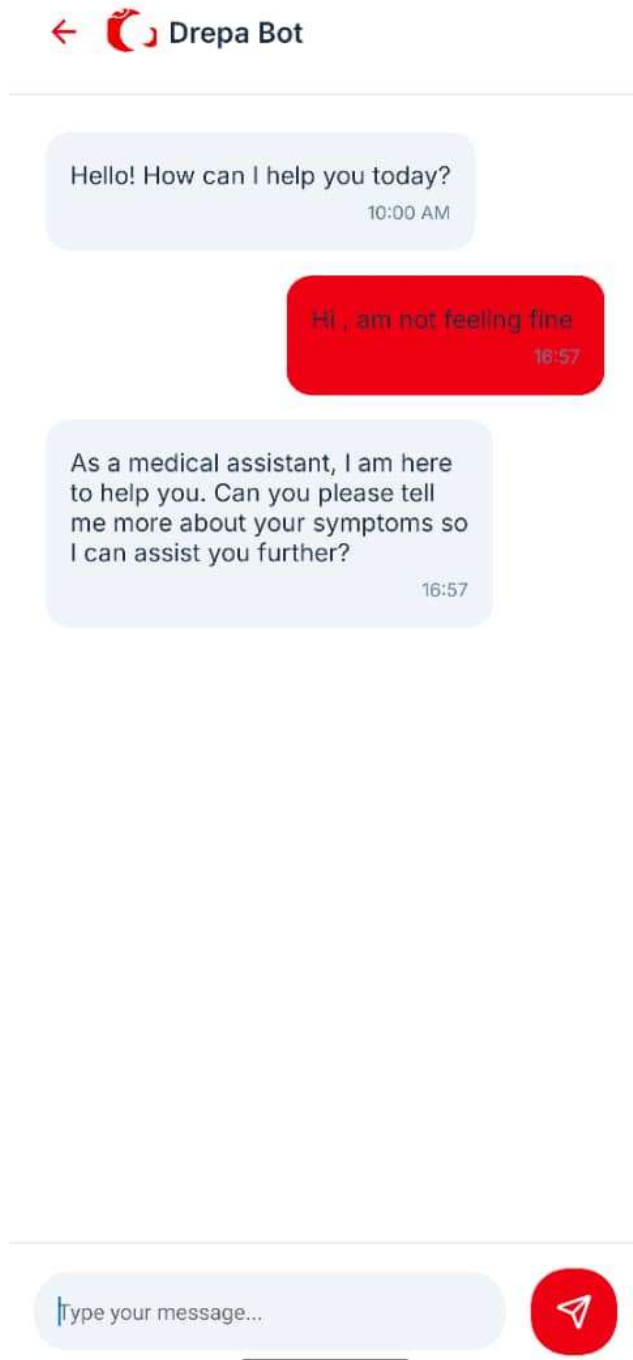


Figure 46 Drepa chatbot screen

CONCLUSION

This user guide has provided a comprehensive overview of our product's features, functionalities, and best practices for optimal use. By following the instructions and tips outlined in this document, you'll be well-equipped to maximize the potential of our product and streamline your workflow.

PERSPECTIVES

Having developed the core functionalities of our application, we plan to improve it further with the following features:

- Integration of a secure payment system.
- User profile management for better personalization.
- Consultation and counseling services with health experts.
- A daily health guide and patient monitoring tools.
- Stronger security measures and provider verification.

GENERAL CONCLUSION

Having come to the end of our project and our internship at REALIZE, we can proudly say that this experience has been both enriching and challenging. Throughout the three months, we were faced with moments of stress, failures, and successes, but each step helped us grow both professionally and personally. Our time at REALIZE was very beneficial as it allowed us to adapt to the professional environment, strengthen our teamwork spirit, and apply the knowledge acquired in class to real-world problems. For our project, we worked on the theme: **“Conception and Realization of a Digital Health Support and Genotype Compatibility Platform for Sickle Cell Prevention in Cameroon.”** Our greatest desire is to contribute to reducing the early death rate caused by sickle cell disease by equipping families with accessible tools for awareness, genotype risk assessment, and preventive guidance. We began by identifying the existing challenges in the current system for sickle cell healthcare in Cameroon and formalized them in our specification book. We then carried out an in-depth analysis using UML and 2TUP methodology and modeled our system with the help of Visual Paradigm. For the realization, we made use of **React Native** for the mobile front-end, **Node.js** for the backend, and **MongoDB** as our database, ensuring a reliable and scalable architecture. The result of our work, Drepalife, will allow families to easily access genotype compatibility information, receive educational support, consult verified healthcare providers, and benefit from an integrated chatbot assistant for guidance. We will not stop here. In the near future, important updates and improvements will be made in terms of functionalities and security in order to make the system more reliable and impactful for communities across Cameroon and, hopefully, beyond.



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